



Electrical Machine I

Course Code: EEE 2105

Transformer

Md. Nuhi-Alamin

Assistant Professor

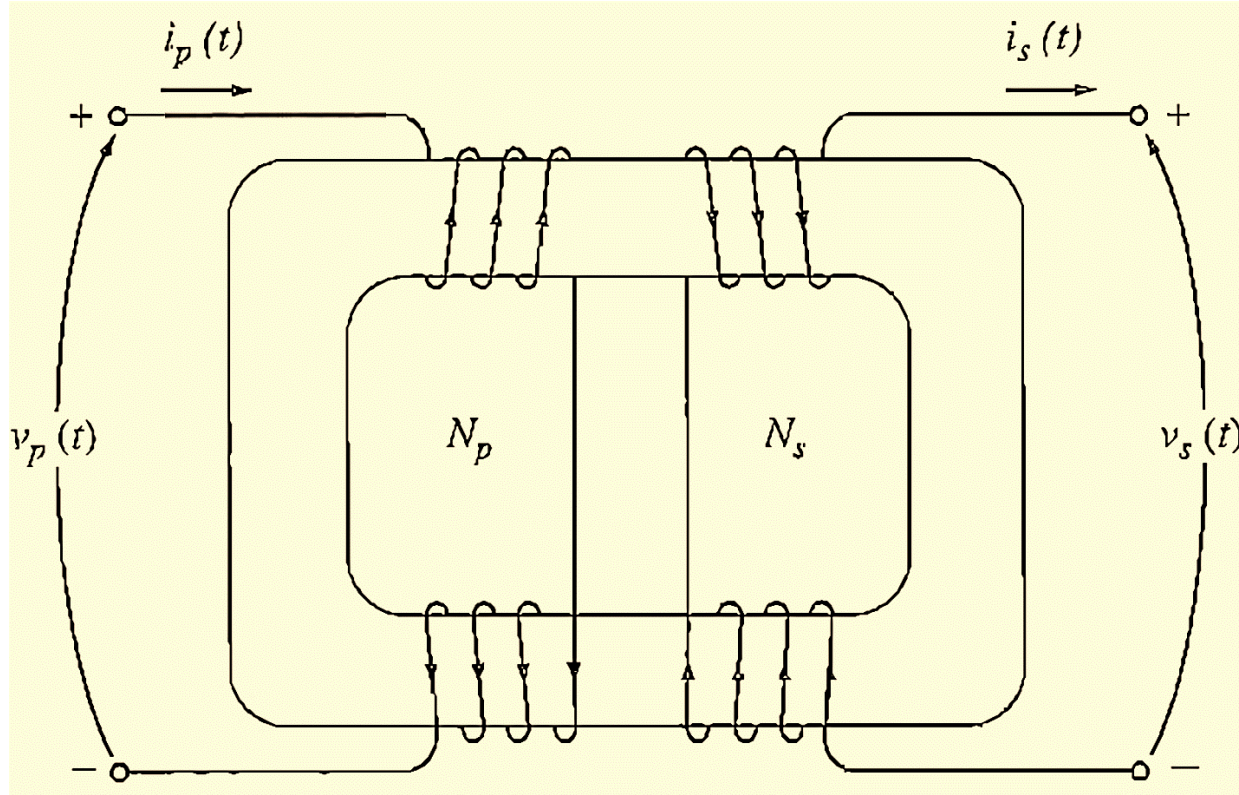
Dept. of Electrical & Electronic Engineering

RUET

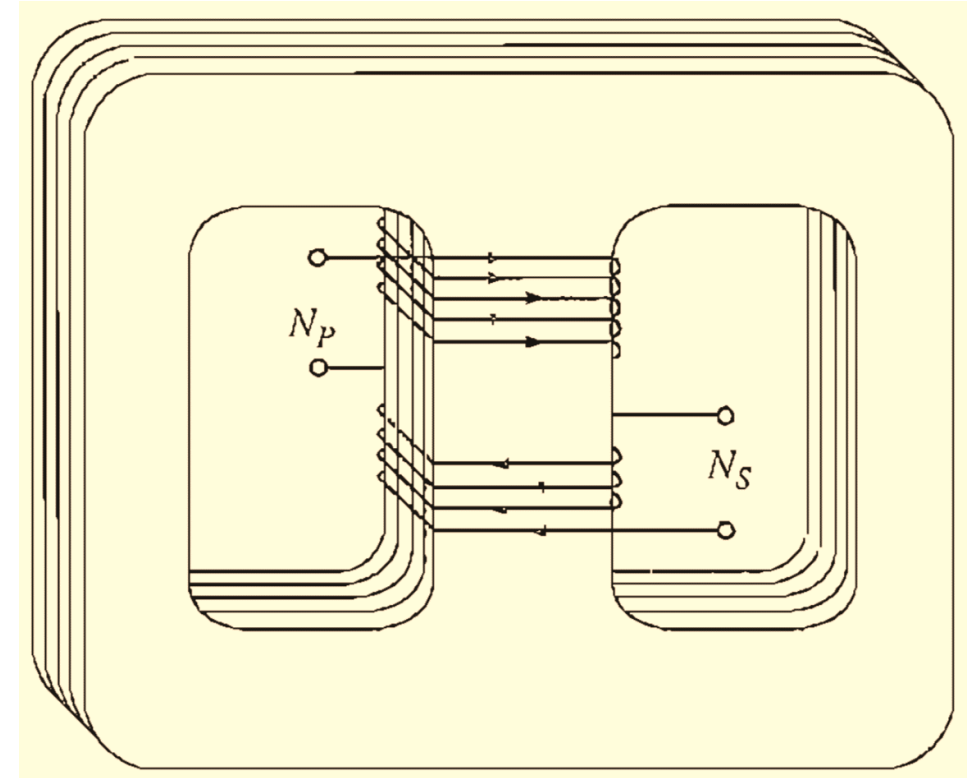
E-mail: nuhialamin@eee.ruet.ac.bd

Cell: 01708-917724

Construction and Types



Core Type Construction



Shell Type Construction

The primary and secondary windings in a physical transformer are wrapped one on top of the other with the low-voltage winding innermost. Such an arrangement serves two purposes:

1. It simplifies the problem of insulating the high-voltage winding from the core.
2. It results in much less leakage flux than would be the case if the two windings were separated by a distance on the core.

Unit Transformer: A transformer connected to the output of a generator and used to step its voltage up to transmission levels (110 + kV) is sometimes called a unit transformer.

Substation Transformer: The transformer at the other end of the transmission line, which steps the voltage down from transmission levels to distribution levels (from 2.3 to 34.5 kV), is called a substation transformer.

Distribution Transformer: The transformer that takes the distribution voltage and steps it down to the final voltage at which the power is actually used (110, 208, 220 V, etc.) is called a distribution transformer.

Potential Transformer: A device specially designed to sample a high voltage and produce a low secondary voltage directly proportional to it is called a potential transformer. A power transformer also produces a secondary voltage directly proportional to its primary voltage; **the difference between a potential transformer and a power transformer is that the potential transformer is designed to handle only a very small current.**

Current Transformer: A device designed to provide a secondary current much smaller than but directly proportional to its primary current is called a current transformer.

Ideal Transformer

An ideal transformer is a lossless device with an input winding and an output winding.

$$\frac{v_p(t)}{v_s(t)} = \frac{N_p}{N_s} = a$$

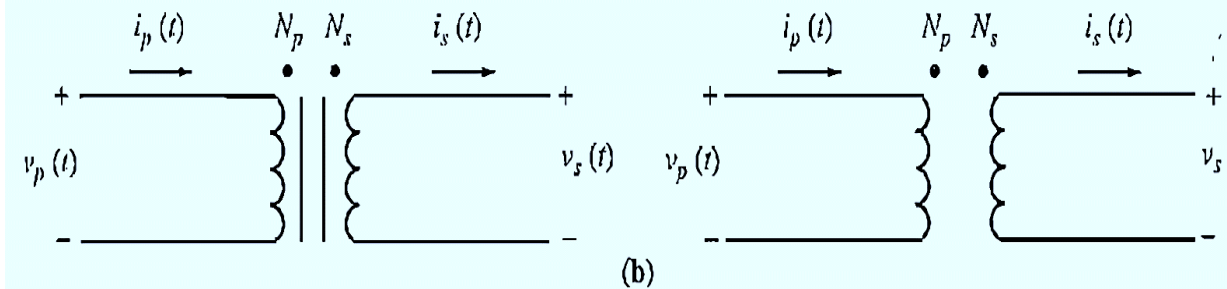
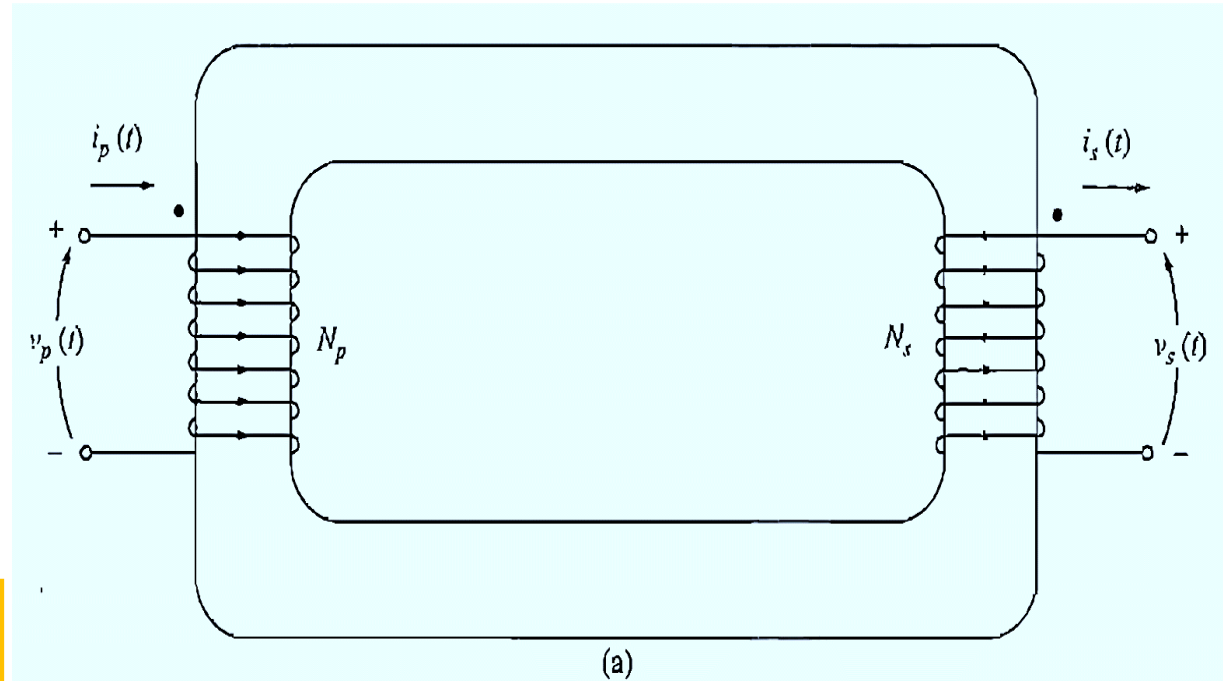
where a is defined to be the *turns ratio* of the transformer:

$$N_p i_p(t) = N_s i_s(t)$$

$$\frac{i_p(t)}{i_s(t)} = \frac{1}{a}$$

If the primary voltage is positive at the dotted end of the winding with respect to the undotted end, then the secondary voltage will be positive at the dotted end also.

If the primary current of the transformer flows into the dotted end of the primary winding, the secondary current will flow out of the dotted end of the secondary winding.



Power in an Ideal Transformer

$$P_{\text{out}} = V_S I_S \cos \theta$$

Applying the turns-ratio equations gives $V_S = V_P/a$ and $I_S = aI_P$, so

$$P_{\text{out}} = \frac{V_P}{a} (aI_P) \cos \theta$$

$$P_{\text{out}} = V_P I_P \cos \theta = P_{\text{in}}$$

Thus, *the output power of an ideal transformer is equal to its input power.*

The same relationship applies to reactive power Q and apparent power S :

$$Q_{\text{in}} = V_P I_P \sin \theta = V_S I_S \sin \theta = Q_{\text{out}}$$

$$S_{\text{in}} = V_P I_P = V_S I_S = S_{\text{out}}$$

Impedance Transformation through a Transformer

$$Z_L = \frac{\mathbf{V}_L}{\mathbf{I}_L}$$

The apparent impedance of the primary circuit of the transformer is

$$Z'_L = \frac{\mathbf{V}_P}{\mathbf{I}_P}$$

Since the primary voltage can be expressed as

$$\mathbf{V}_P = a\mathbf{V}_S$$

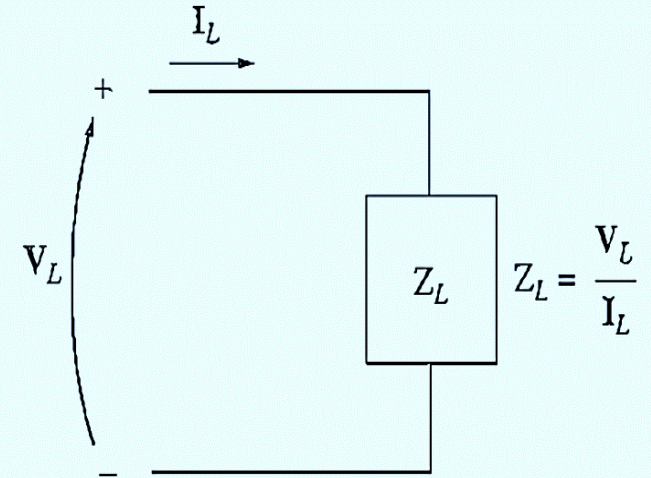
and the primary current can be expressed as

$$\mathbf{I}_P = \frac{\mathbf{I}_S}{a}$$

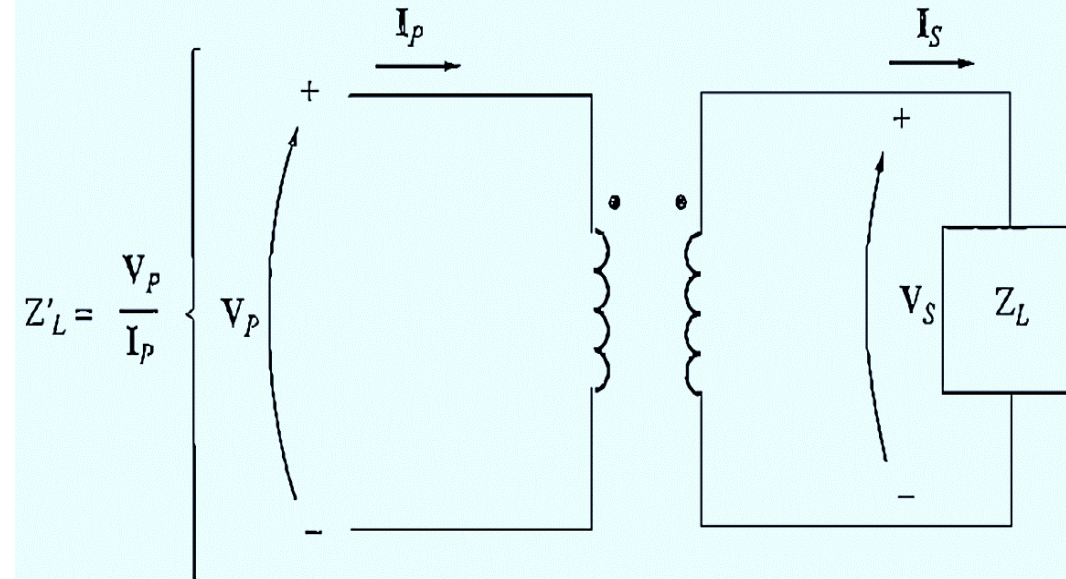
the apparent impedance of the primary is

$$Z'_L = \frac{\mathbf{V}_P}{\mathbf{I}_P} = \frac{a\mathbf{V}_S}{\mathbf{I}_S/a} = a^2 \frac{\mathbf{V}_S}{\mathbf{I}_S}$$

$$\boxed{Z'_L = a^2 Z_L}$$



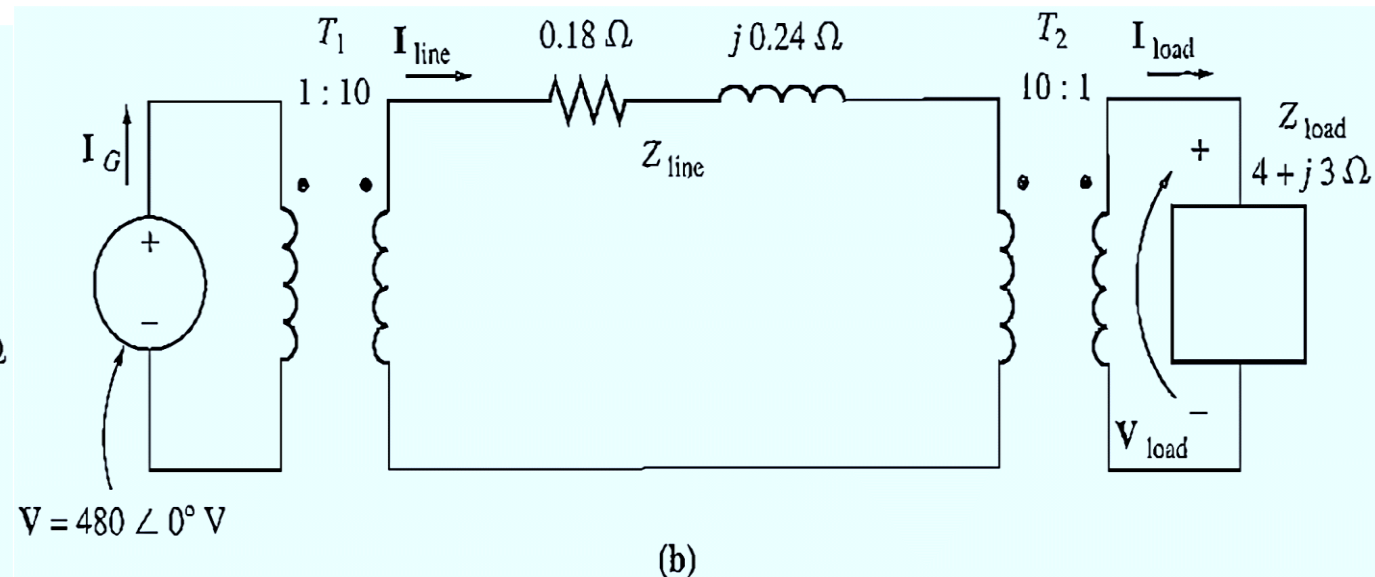
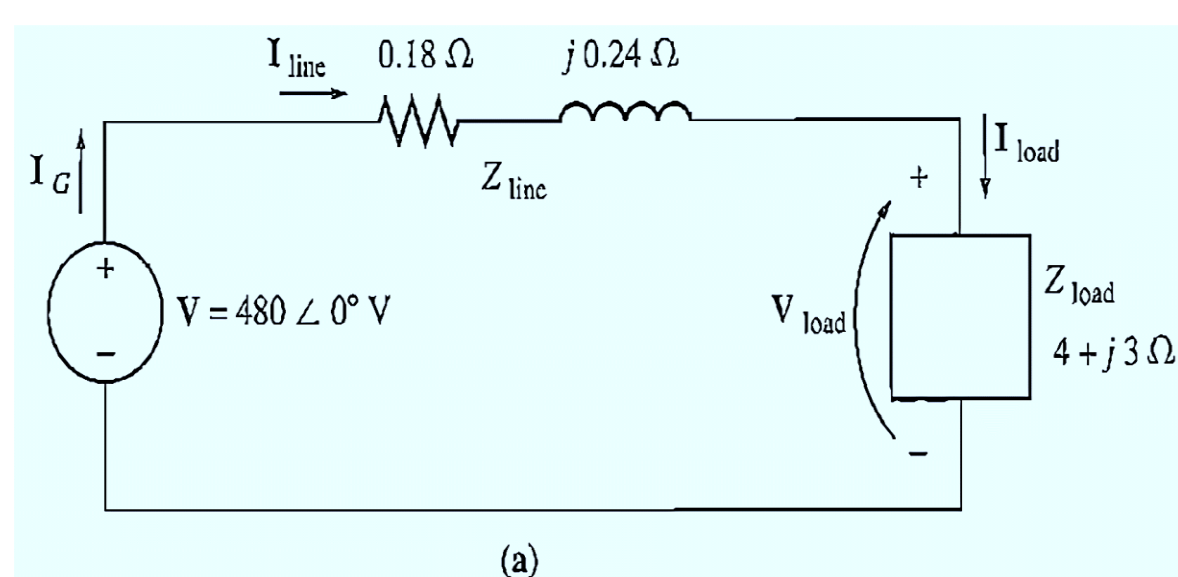
(a)



(b)

Example 2–1. A single-phase power system consists of a 480-V 60-Hz generator supplying a load $Z_{\text{load}} = 4 + j3 \Omega$ through a transmission line of impedance $Z_{\text{line}} = 0.18 + j0.24 \Omega$. Answer the following questions about this system.

- (a) If the power system is exactly as described above (and shown in Figure 2–6a), what will the voltage at the load be? What will the transmission line losses be?
- (b) Suppose a 1:10 step-up transformer is placed at the generator end of the transmission line and a 10:1 step-down transformer is placed at the load end of the line (as shown in Figure 2–6b). What will the load voltage be now? What will the transmission line losses be now?



Solution

(a) Figure 2–6a shows the power system without transformers. Here $\mathbf{I}_G = \mathbf{I}_{\text{line}} = \mathbf{I}_{\text{load}}$. The line current in this system is given by

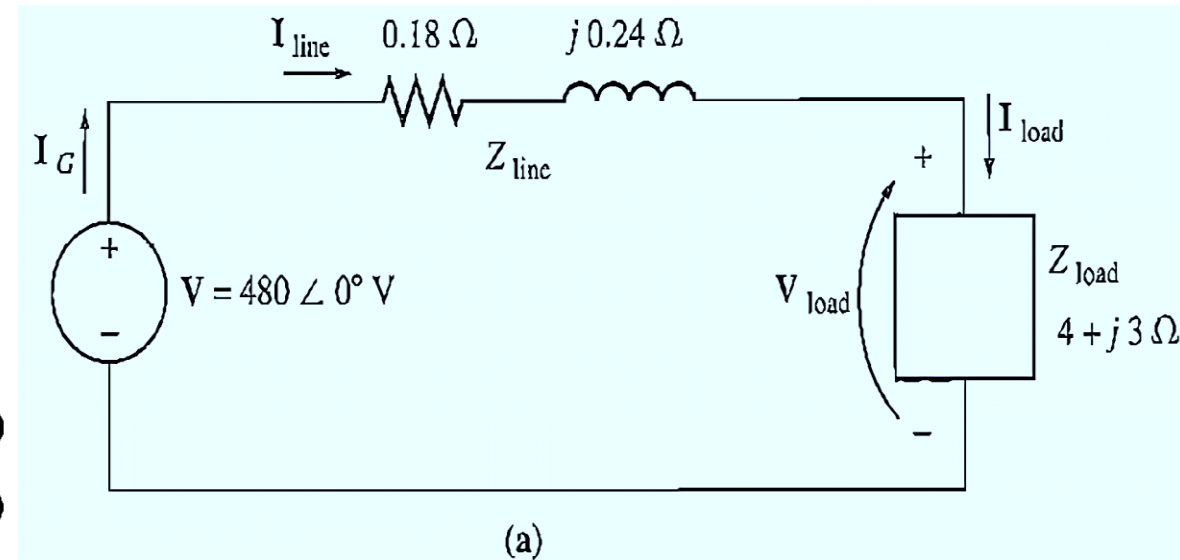
$$\begin{aligned}\mathbf{I}_{\text{line}} &= \frac{\mathbf{V}}{\mathbf{Z}_{\text{line}} + \mathbf{Z}_{\text{load}}} \\ &= \frac{480 \angle 0^\circ \text{ V}}{(0.18 \Omega + j0.24 \Omega) + (4 \Omega + j3 \Omega)} \\ &= \frac{480 \angle 0^\circ}{4.18 + j3.24} = \frac{480 \angle 0^\circ}{5.29 \angle 37.8^\circ} \\ &= 90.8 \angle -37.8^\circ \text{ A}\end{aligned}$$

Therefore the load voltage is

$$\begin{aligned}\mathbf{V}_{\text{load}} &= \mathbf{I}_{\text{line}} \mathbf{Z}_{\text{load}} \\ &= (90.8 \angle -37.8^\circ \text{ A})(4 \Omega + j3 \Omega) \\ &= (90.8 \angle -37.8^\circ \text{ A})(5 \angle 36.9^\circ \Omega) \\ &= 454 \angle -0.9^\circ \text{ V}\end{aligned}$$

and the line losses are

$$\begin{aligned}P_{\text{loss}} &= (I_{\text{line}})^2 R_{\text{line}} \\ &= (90.8 \text{ A})^2 (0.18 \Omega) = 1484 \text{ W}\end{aligned}$$



(b) Figure 2–6b shows the power system with the transformers. To analyze this system, it is necessary to convert it to a common voltage level. This is done in two steps:

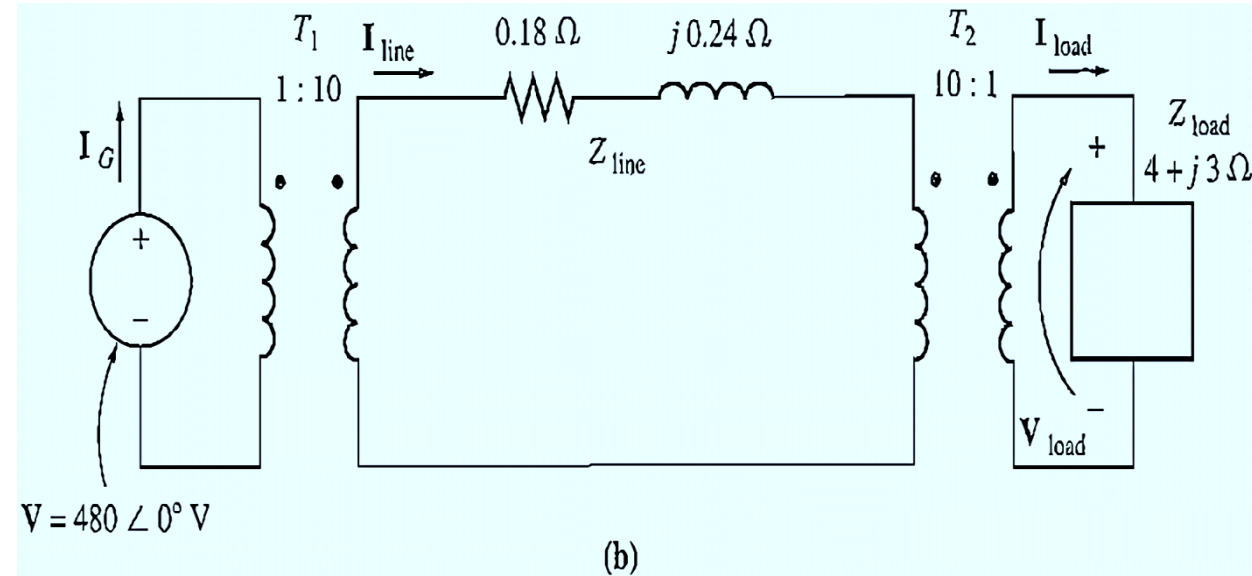
1. Eliminate transformer T_2 by referring the load over to the transmission line's voltage level.
2. Eliminate transformer T_1 by referring the transmission line's elements and the equivalent load at the transmission line's voltage over to the source side.

The value of the load's impedance when reflected to the transmission system's voltage is

$$\begin{aligned} Z'_{\text{load}} &= a^2 Z_{\text{load}} \\ &= \left(\frac{10}{1}\right)^2 (4 \Omega + j3 \Omega) \\ &= 400 \Omega + j300 \Omega \end{aligned}$$

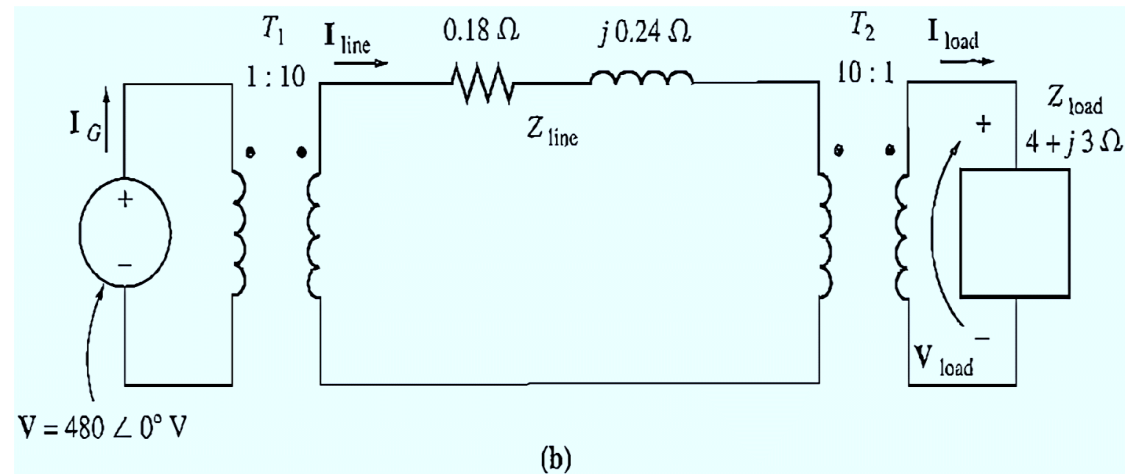
The total impedance at the transmission line level is now

$$\begin{aligned} Z_{\text{eq}} &= Z_{\text{line}} + Z'_{\text{load}} \\ &= 400.18 + j300.24 \Omega = 500.3 \angle 36.88^\circ \Omega \end{aligned}$$



This equivalent circuit is shown in Figure 2–7a. The total impedance at the transmission line level ($Z_{\text{line}} + Z'_{\text{load}}$) is now reflected across T_1 to the source's voltage level:

$$\begin{aligned}
 Z'_{\text{eq}} &= a^2 Z_{\text{eq}} \\
 &= a^2 (Z_{\text{line}} + Z'_{\text{load}}) \\
 &= \left(\frac{1}{10}\right)^2 (0.18 \Omega + j0.24 \Omega + 400 \Omega + j300 \Omega) \\
 &= (0.0018 \Omega + j0.0024 \Omega + 4 \Omega + j3 \Omega) \\
 &= 5.003 \angle 36.88^\circ \Omega
 \end{aligned}$$

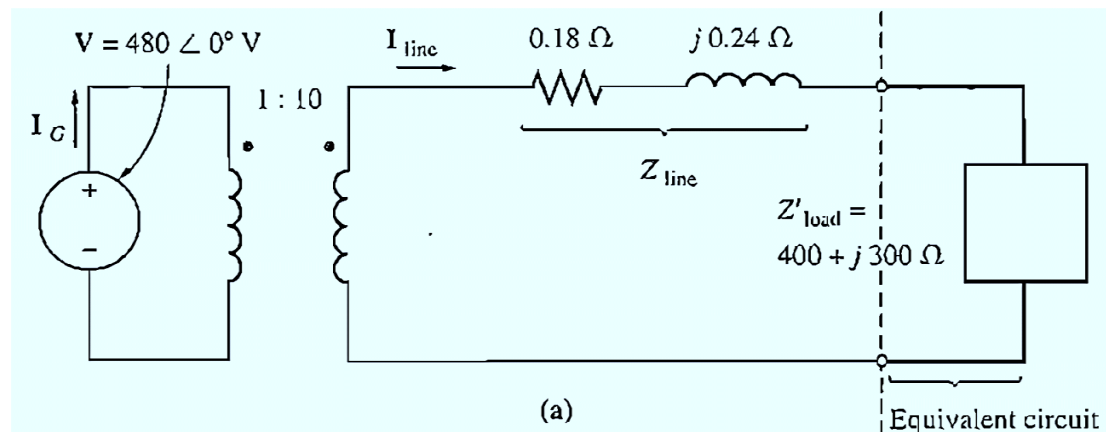


Notice that $Z'_{\text{load}} = 4 + j3 \Omega$ and $Z'_{\text{line}} = 0.0018 + j0.0024 \Omega$. The resulting equivalent circuit is shown in Figure 2–7b. The generator's current is

$$I_G = \frac{480 \angle 0^\circ \text{ V}}{5.003 \angle 36.88^\circ \Omega} = 95.94 \angle -36.88^\circ \text{ A}$$

Knowing the current I_G , we can now work back and find I_{line} and I_{load} . Working back through T_1 , we get

$$\begin{aligned}
 N_{P1} I_G &= N_{S1} I_{\text{line}} \\
 I_{\text{line}} &= \frac{N_{P1}}{N_{S1}} I_G \\
 &= \frac{1}{10} (95.94 \angle -36.88^\circ \text{ A}) = 9.594 \angle -36.88^\circ \text{ A}
 \end{aligned}$$



Working back through T_2 gives

$$N_{P2} \mathbf{I}_{\text{line}} = N_{S2} \mathbf{I}_{\text{load}}$$

$$\mathbf{I}_{\text{load}} = \frac{N_{P2}}{N_{S2}} \mathbf{I}_{\text{line}}$$

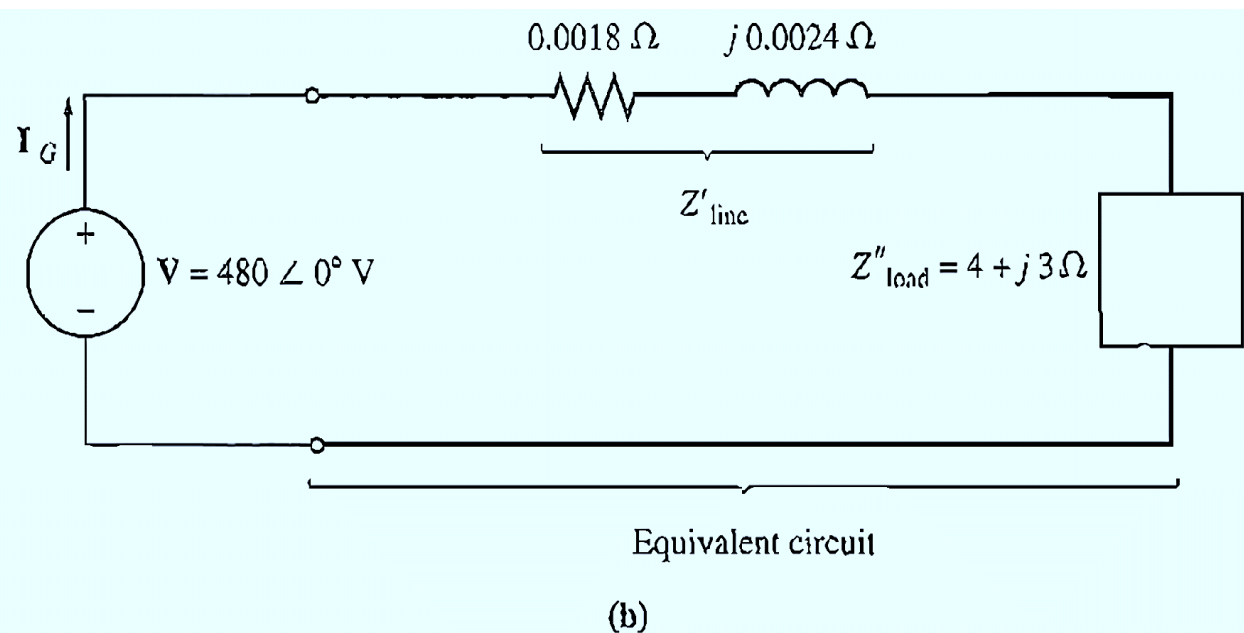
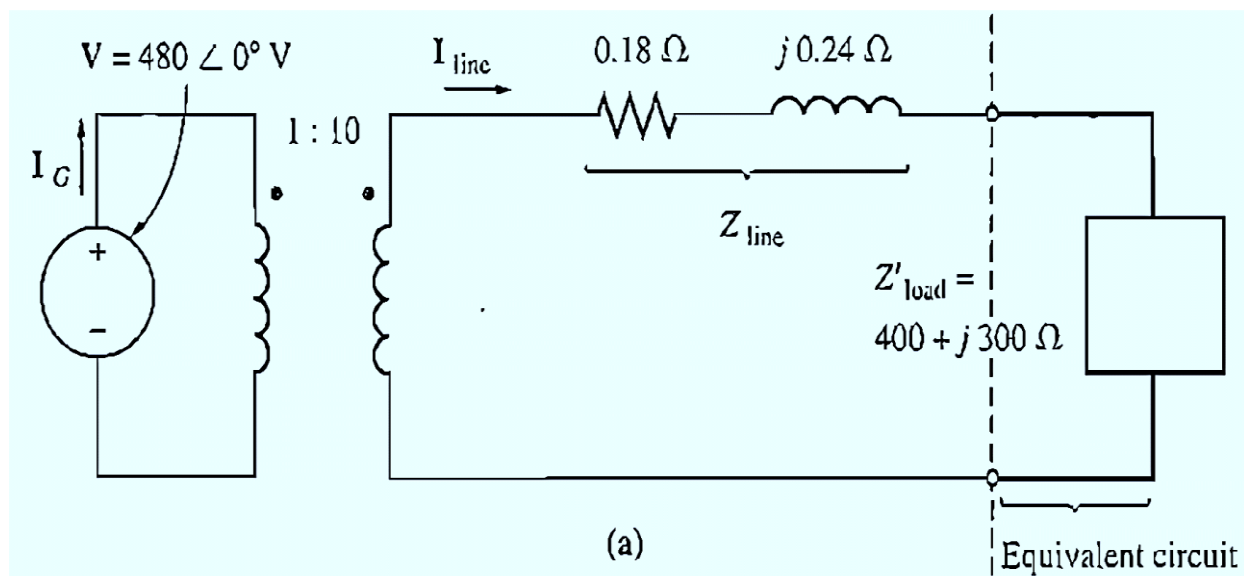
$$= \frac{10}{1} (9.594 \angle -36.88^\circ \text{ A}) = 95.94 \angle -36.88^\circ \text{ A}$$

and the line losses are given by

$$\begin{aligned} P_{\text{loss}} &= (I_{\text{line}})^2 R_{\text{line}} \\ &= (9.594 \text{ A})^2 (0.18 \Omega) = 16.7 \text{ W} \end{aligned}$$

It is now possible to answer the questions originally asked. The load voltage is given by

$$\begin{aligned} \mathbf{V}_{\text{load}} &= \mathbf{I}_{\text{load}} \mathbf{Z}_{\text{load}} \\ &= (95.94 \angle -36.88^\circ \text{ A})(5 \angle 36.87^\circ \Omega) \\ &= 479.7 \angle -0.01^\circ \text{ V} \end{aligned}$$



Losses in a Transformer

1. *Copper (I^2R) losses.* Copper losses are the resistive heating losses *in the primary and secondary windings* of the transformer. They are proportional to the square of the current in the windings.
2. *Eddy current losses.* Eddy current losses are resistive heating losses *in the core* of the transformer. They are proportional to the square of the voltage applied to the transformer.
3. *Hysteresis losses.* Hysteresis losses are associated with the rearrangement of the magnetic domains in the core during each half-cycle, as explained in Chapter 1. They are a complex, nonlinear function of the voltage applied to the transformer.
4. *Leakage flux.* The fluxes ϕ_{LP} and ϕ_{LS} which escape the core and pass through only one of the transformer windings are leakage fluxes. These escaped fluxes produce a *leakage inductance* in the primary and secondary coils, and the effects of this inductance must be accounted for.

E.M.F. Equation of a Transformer

$$\begin{aligned}\Phi_m &= \text{Maximum flux in core in webers} \\ &= B_m \times A\end{aligned}$$

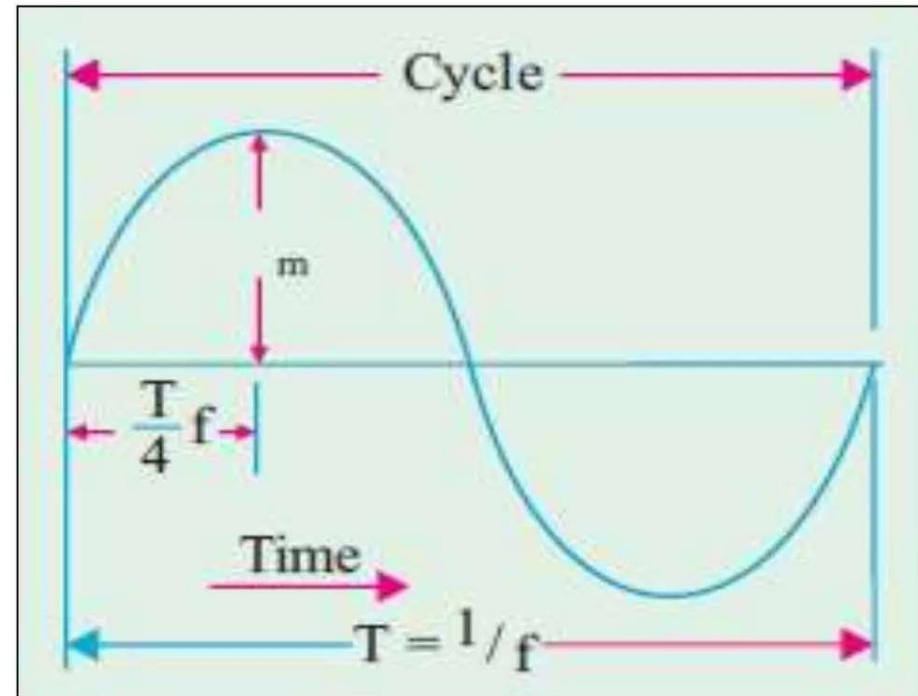
$$\begin{aligned}\text{Average rate of change of flux} &= \frac{\Phi_m}{1/4f} \\ &= 4f\Phi_m \text{ Wb/s or volt}\end{aligned}$$

$$\text{r.m.s. value of e.m.f./turn} = 1.11 \times 4f\Phi_m = 4.44f\Phi_m \text{ volt}$$

$$E_1 = 4.44fN_1\Phi_m = 4.44fN_1B_mA$$

$$E_2 = 4.44fN_2\Phi_m = 4.44fN_2B_mA$$

$$E_1/N_1 = E_2/N_2 = 4.44f\Phi_m.$$



32.7 Voltage Transformation Ratio (K)

From equations (i) and (ii), we get

$$\frac{E_2}{E_1} = \frac{N_2}{N_1} = K$$

This constant K is known as voltage transformation ratio.

(i) If $N_2 > N_1$ i.e. $K > 1$, then transformer is called **step-up** transformer.

(ii) If $N_2 < N_1$ i.e. $K < 1$, then transformer is known as **step-down** transformer.

Again, for an *ideal* transformer, input VA = output VA.

$$V_1 I_1 = V_2 I_2 \text{ or } \frac{I_2}{I_1} = \frac{V_1}{V_2} = \frac{1}{K}$$

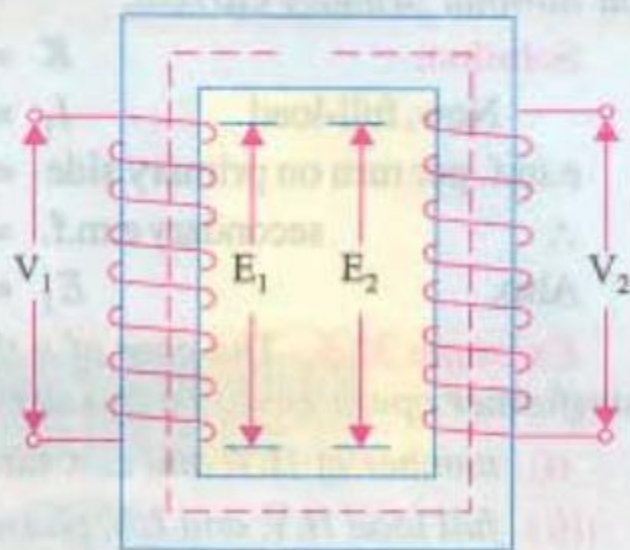


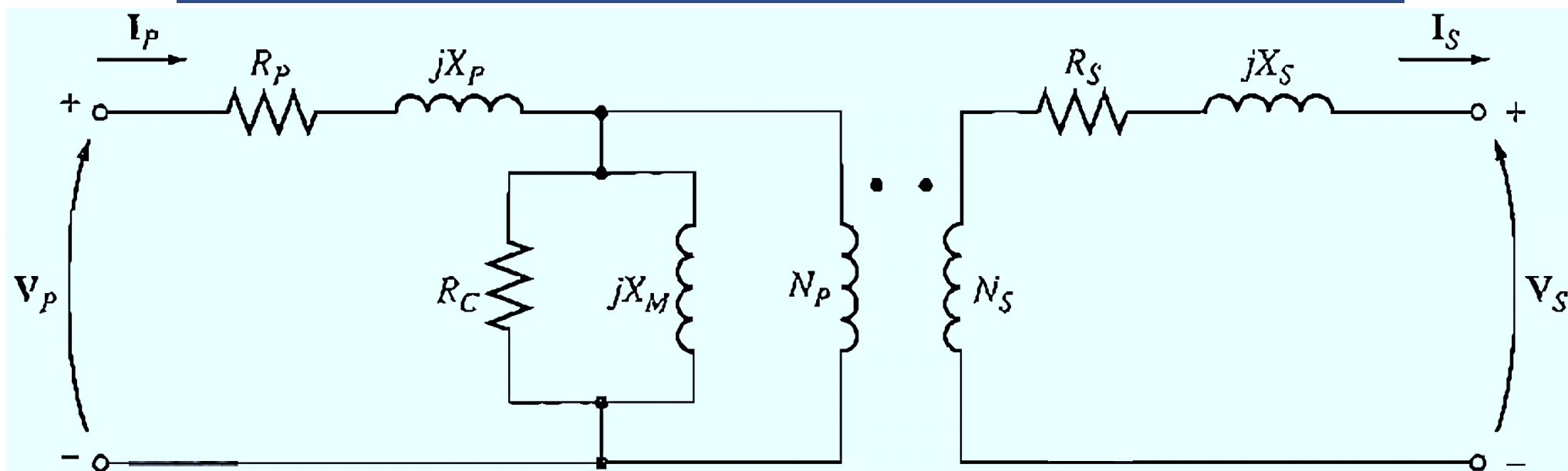
Fig. 32.15

Example 32.1. The maximum flux density in the core of a 250/3000-volts, 50-Hz single-phase transformer is 1.2 Wb/m^2 . If the e.m.f. per turn is 8 volt, determine
(i) primary and secondary turns
(ii) area of the core.

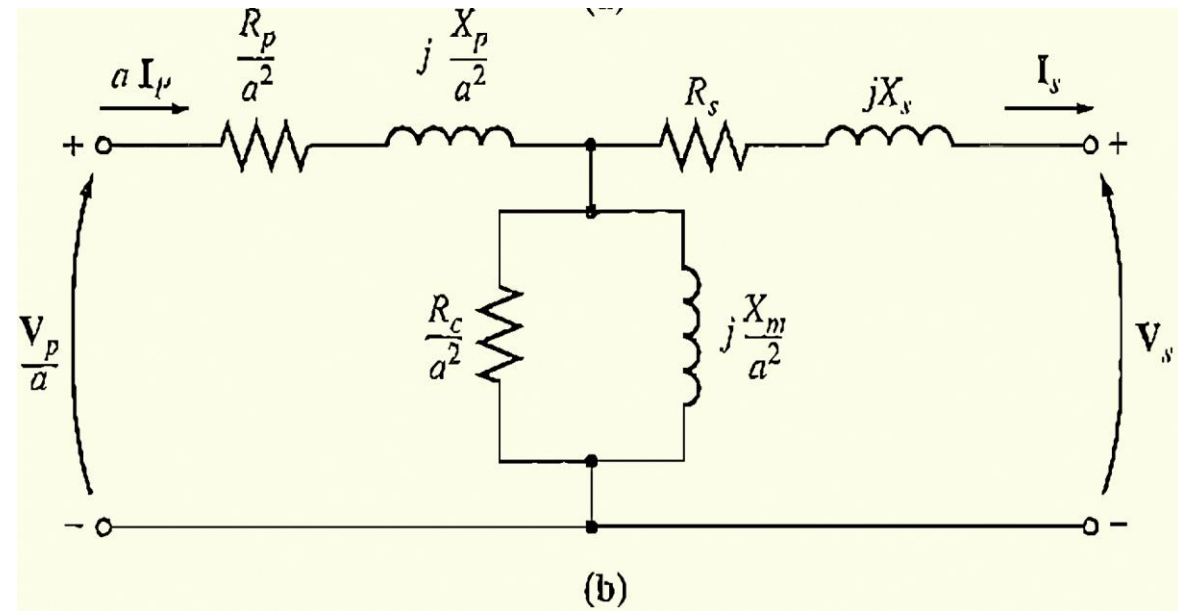
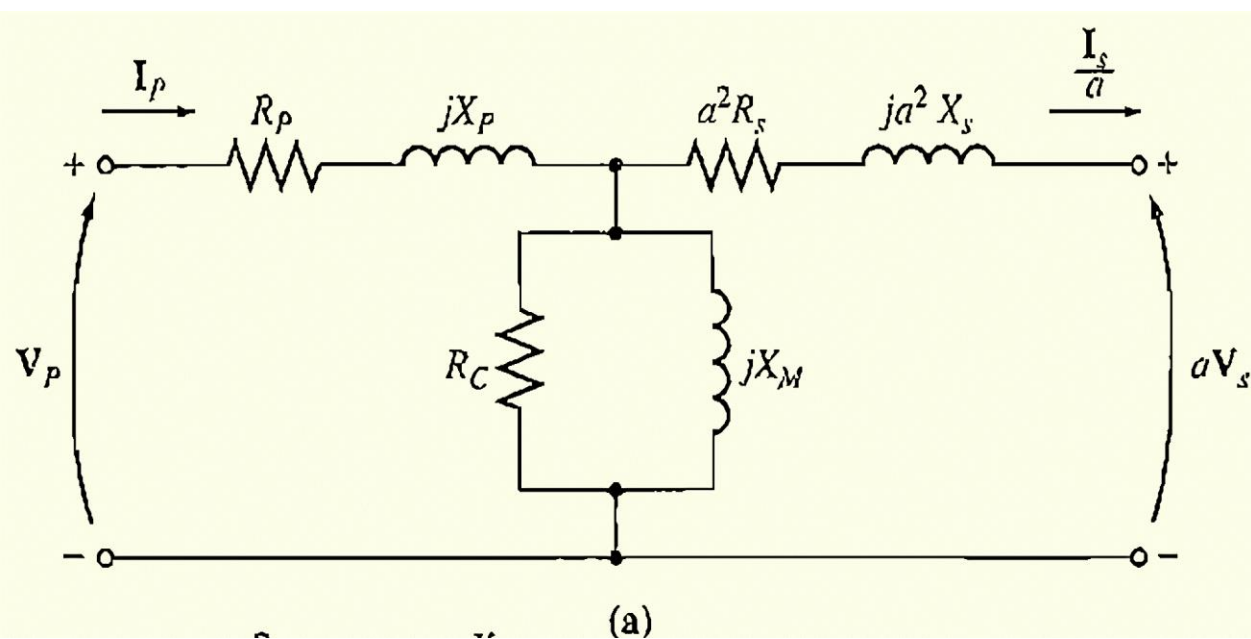
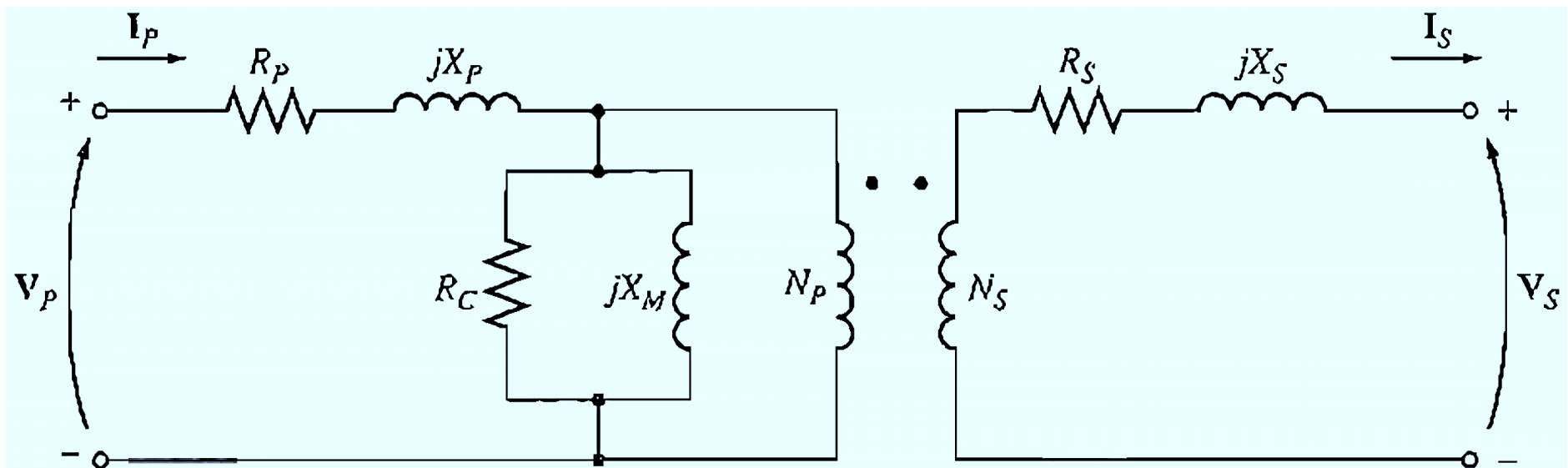
Math: 32.1-32.7

Book : BL Thereja,
Chapter: 32

Exact Equivalent Circuit of a Transformer

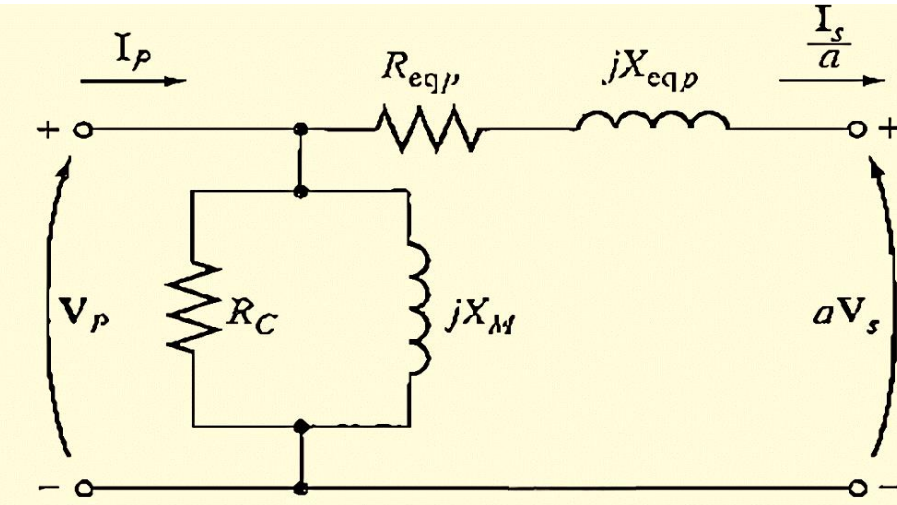


- The magnetization current $\mathbf{i}_m$ is a current proportional (in the unsaturated region) to the voltage applied to the core and lagging the applied voltage by 90° , so it can be modeled by a reactance $\mathbf{X}_M$ connected across the primary voltage source.
- The core-loss current $\mathbf{i}_{h+e}$ is a current proportional to the voltage applied to the core that is in phase with the applied voltage, so it can be modeled by a resistance R_c connected across the primary voltage source.
- R_p is the resistance of the primary winding, $\mathbf{X}_p (= \omega \mathbf{L}_p)$ is the reactance due to the primary leakage inductance, R_s is the resistance of the secondary winding, and $\mathbf{X}_s (= \omega \mathbf{L}_s)$ is the reactance due to the secondary leakage inductance. The excitation branch is modeled by the resistance $\mathbf{R}_c$ (hysteresis and core losses) in parallel with the reactance $\mathbf{X}_M$ (the magnetization current).



(a) The transformer model referred to its primary voltage level. (b) The transformer model referred to its secondary voltage level.

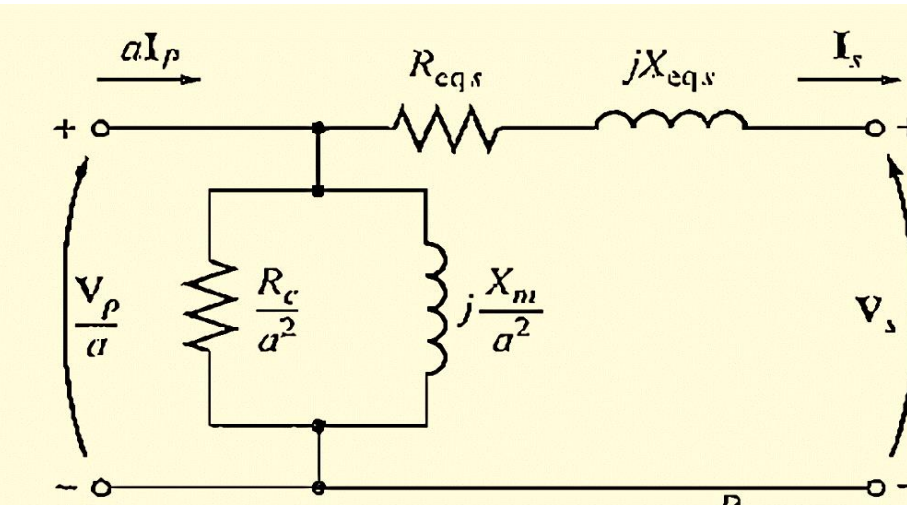
Approximate Equivalent Circuit of a Transformer



(a)

$$R_{eqp} = R_p + a^2 R_s$$

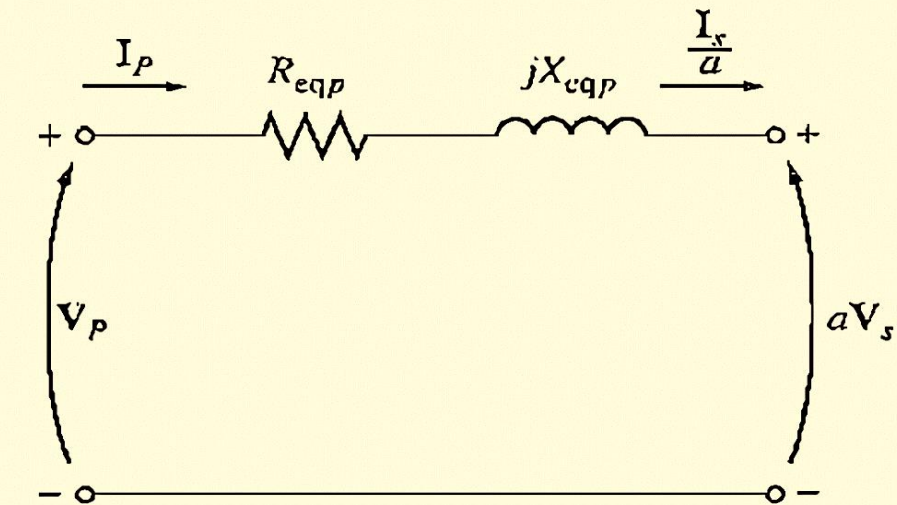
$$X_{eqp} = X_p + a^2 X_s$$



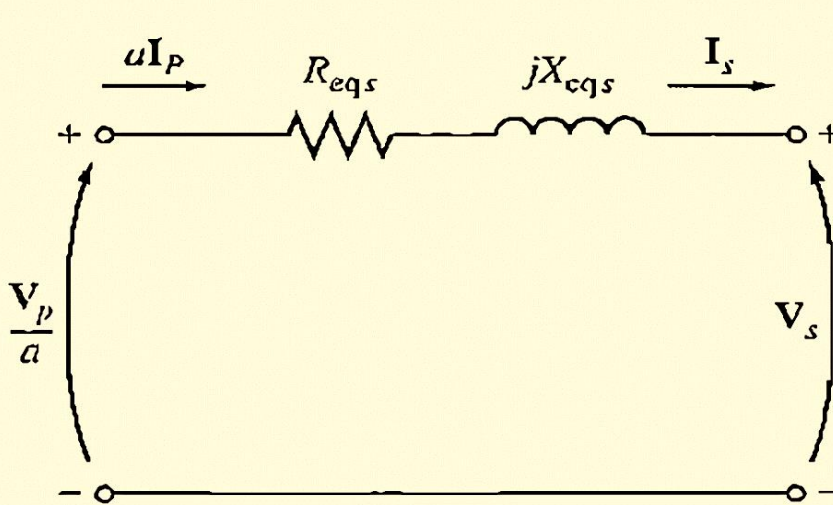
(b)

$$R_{eqs} = \frac{R_p}{a^2} + R_s$$

$$X_{eqs} = \frac{X_p}{a^2} + X_s$$



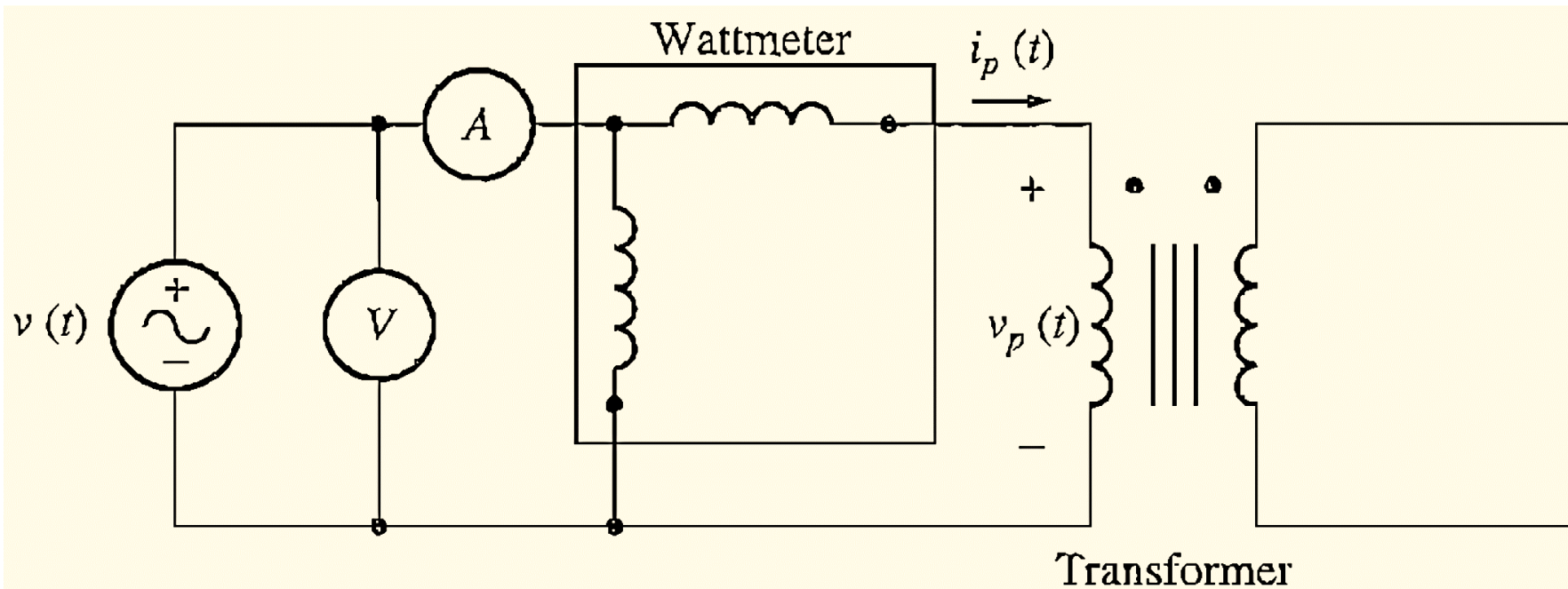
(c)



(d)

Approximate transformer models. (a) Referred to the primary side; (b) referred to the secondary side; (c) with no excitation branch, referred to the primary side; (d) with no excitation branch, referred to the secondary side.

Open Circuit Test of Transformer: Determining Excitation Branch Parameters



All the input current must be flowing through the excitation branch of the transformer. The series elements, R_p and X_p are too small in comparison to R_C and X_M to cause a significant voltage drop, so essentially all the input voltage is dropped across the excitation branch. **Normally done on low voltage side.**

$$G_C = \frac{1}{R_C} \quad B_M = \frac{1}{X_M}$$

$$Y_E = G_C - jB_M$$

$$Y_E = \frac{1}{R_C} - j\frac{1}{X_M}$$

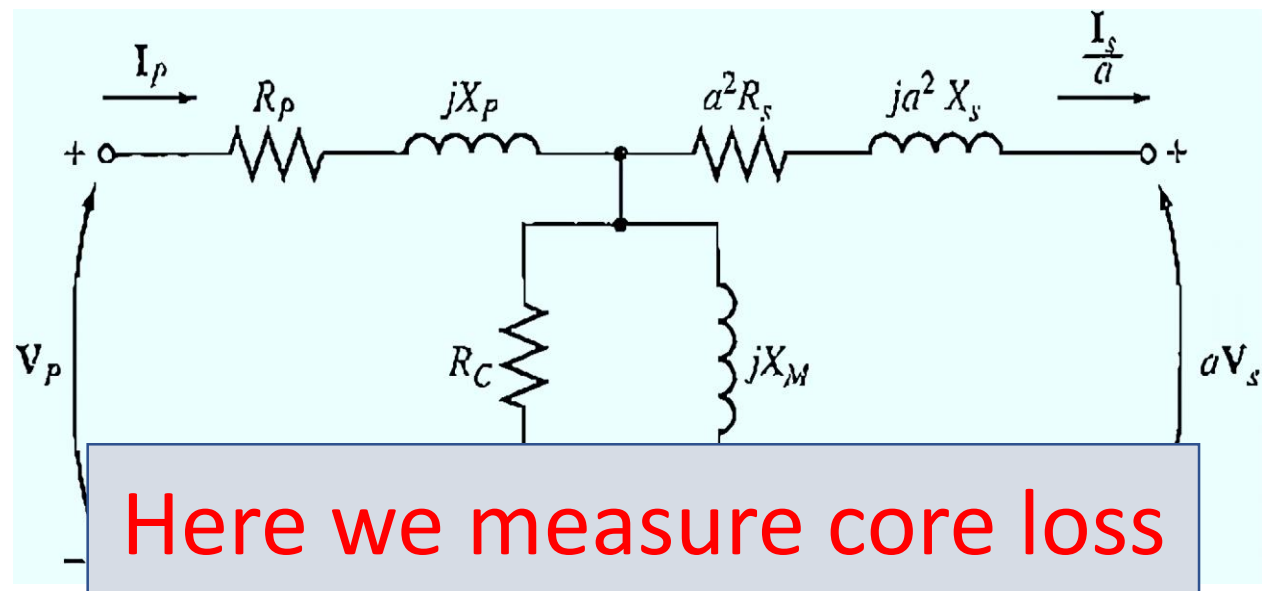
$$|Y_E| = \frac{I_{OC}}{V_{OC}}$$

$$PF = \cos \theta = \frac{P_{OC}}{V_{OC} I_{OC}}$$

$$\theta = \cos^{-1} \frac{P_{OC}}{V_{OC} I_{OC}}$$

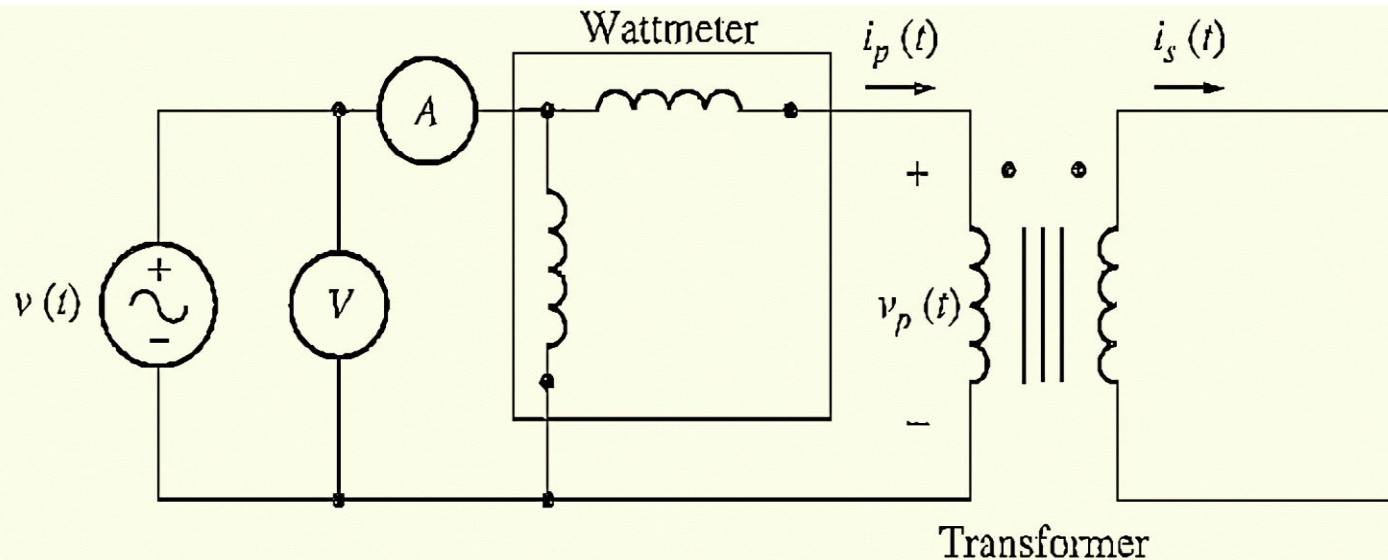
$$Y_E = \frac{I_{OC}}{V_{OC}} \angle -\theta$$

$$Y_E = \frac{I_{OC}}{V_{OC}} \angle -\cos^{-1} PF$$



Here we measure core loss

Short Circuit Test of Transformer: Determining Series Parameters



In the short-circuit test, the **low-voltage terminals of the transformer are short circuited**, and the high-voltage terminals are connected to a variable voltage source.

Since the input voltage is so low during the short-circuit test, negligible current flows through the excitation branch. If the excitation current is ignored, then **all the voltage drop in the transformer can be attributed to the series elements in the circuit.**

$$|Z_{SE}| = \frac{V_{SC}}{I_{SC}}$$

$$\text{PF} = \cos \theta = \frac{P_{SC}}{V_{SC} I_{SC}}$$

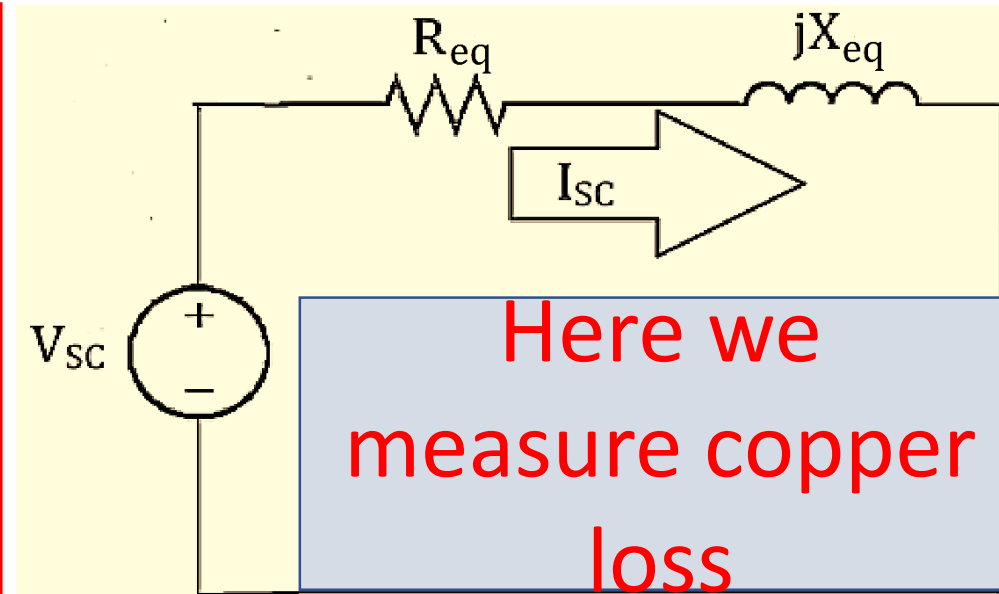
$$\theta = \cos^{-1} \frac{P_{SC}}{V_{SC} I_{SC}}$$

$$Z_{SE} = \frac{V_{SC} \angle 0^\circ}{I_{SC} \angle -\theta^\circ} = \frac{V_{SC}}{I_{SC}} \angle \theta^\circ$$

(Current lags voltage due to the presence of inductance)

$$Z_{SE} = R_{eq} + jX_{eq}$$

$$Z_{SE} = (R_P + a^2 R_S) + j(X_P + a^2 X_S)$$



Example 2-2. The equivalent circuit impedances of a 20-kVA, 8000/240 V, 60-Hz transformer are to be determined. The open-circuit test was performed on the secondary side of the transformer (to reduce the maximum voltage to be measured) and the short circuit test were performed on the primary side of the transformer (to reduce the maximum current to be measured). Find the impedances of the approximate equivalent circuit referred to the primary side, and sketch that circuit. The following data were taken:

Open-circuit test (on secondary)	Short-circuit test (on primary)
$V_{OC} = 240 \text{ V}$	$V_{SC} = 489 \text{ V}$
$I_{OC} = 7.133 \text{ A}$	$I_{SC} = 2.5 \text{ A}$
$P_{OC} = 400 \text{ W}$	$P_{SC} = 240 \text{ W}$

Solution: Power factor during open circuit test:

$$\text{PF} = \cos \theta = \frac{P_{OC}}{V_{OC} I_{OC}}$$

$$\text{PF} = \cos \theta = \frac{400 \text{ W}}{(240 \text{ V})(7.133 \text{ A})}$$

$$\text{PF} = 0.234 \text{ lagging}$$

The excitation admittance is given by

$$Y_E = \frac{I_{OC}}{V_{OC}} \angle -\cos^{-1} \text{PF}$$

$$Y_E = \frac{7.133 \text{ A}}{240 \text{ V}} \angle -\cos^{-1} 0.234$$

$$Y_E = 0.0297 \angle -76.5^\circ \text{ S}$$

$$Y_E = 0.00693 - j 0.02888 = \frac{1}{R_C} - j \frac{1}{X_M}$$

Therefore, the values of the excitation branch *referred to the low-voltage (secondary) side* are

$$R_C = \frac{1}{0.00693} = 144 \Omega$$

$$X_M = \frac{1}{0.02888} = 34.63 \Omega$$

The power factor during the *short-circuit* test is

$$\text{PF} = \cos \theta = \frac{P_{SC}}{V_{SC} I_{SC}}$$

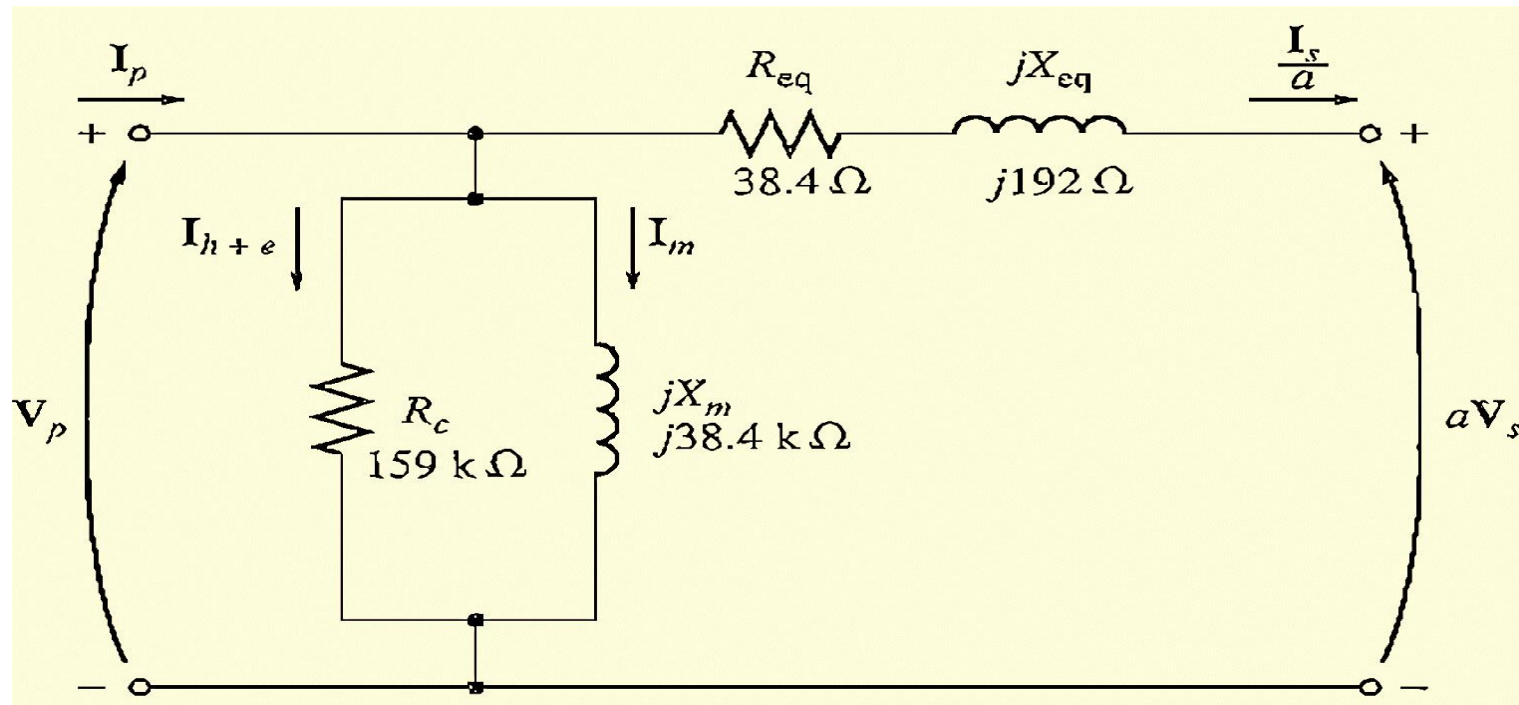
$$\text{PF} = \cos \theta = \frac{240 \text{ W}}{(489 \text{ V})(2.5 \text{ A})} = 0.196 \text{ lagging}$$

The series impedance is given by

$$Z_{SE} = \frac{V_{SC}}{I_{SC}} \angle \cos^{-1} \text{PF}$$

$$Z_{SE} = \frac{489 \text{ V}}{2.5 \text{ A}} \angle 78.7^\circ$$

$$Z_{SE} = 195.6 \angle 78.7^\circ = 38.4 + j192 \Omega$$



Therefore, the equivalent resistance and reactance *referred to the high-voltage (primary) side* are

$$R_{eq} = 38.4 \Omega \quad X_{eq} = 192 \Omega$$

The resulting simplified equivalent circuit referred to the high-voltage (primary) side can be found by converting the excitation branch values to the high-voltage side.

$$R_{C,P} = a^2 R_{C,S} = (33.333)^2 (144 \Omega) = 159 \text{ k}\Omega$$

$$X_{M,P} = a^2 X_{M,S} = (33.333)^2 (34.63 \Omega) = 38.4 \text{ k}\Omega$$

Example. A 5 kVA 200/1000 V, 50 Hz, single-phase transformer gave the following test results:

O.C. Test (L.V. Side): 2000 V, 1.2 A, 90 W

S.C. Test (H.V. Side): 50 V, 5 A, 110 W

(i) Calculate the parameters of the equivalent circuit referred to the L.V. side.

Transformer Voltage Regulation

Because a real transformer has series impedances within it, the output voltage of a transformer varies with the load even if the input voltage remains constant.

Full-load voltage regulation is a quantity that compares the output voltage of the transformer at no load with the output voltage at full load. It is defined by the equation

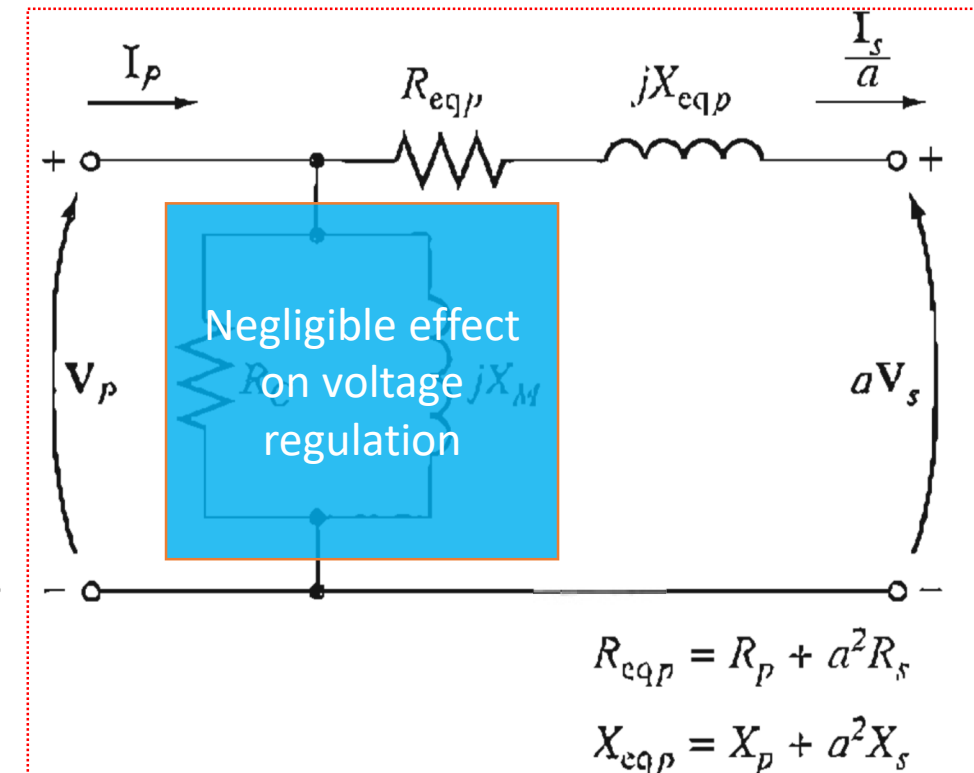
$$VR = \frac{V_{S,nl} - V_{S,fl}}{V_{S,fl}} \times 100\%$$

Since at no load, $V_S = V_P/a$, the voltage regulation can also be expressed as

$$VR = \frac{V_P/a - V_{S,fl}}{V_{S,fl}} \times 100\%$$

If the transformer equivalent circuit is in the per-unit system, then voltage regulation can be expressed as

$$VR = \frac{V_{P,pu} - V_{S,fl,pu}}{V_{S,fl,pu}} \times 100\%$$

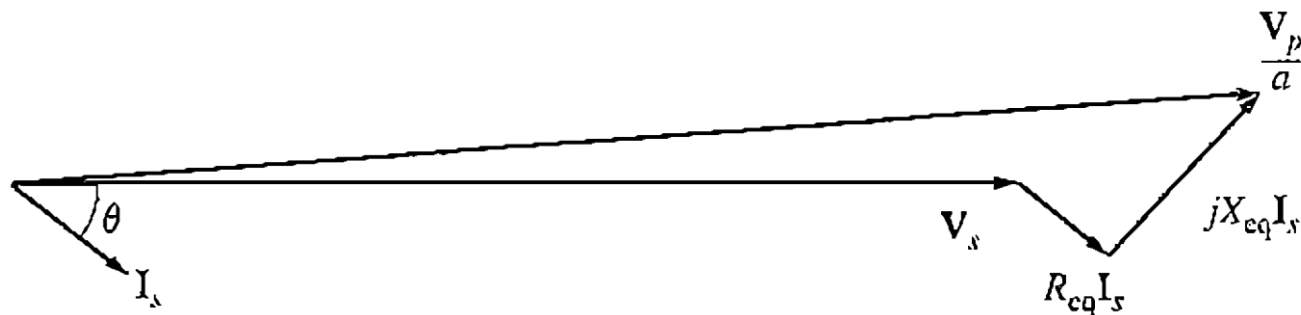
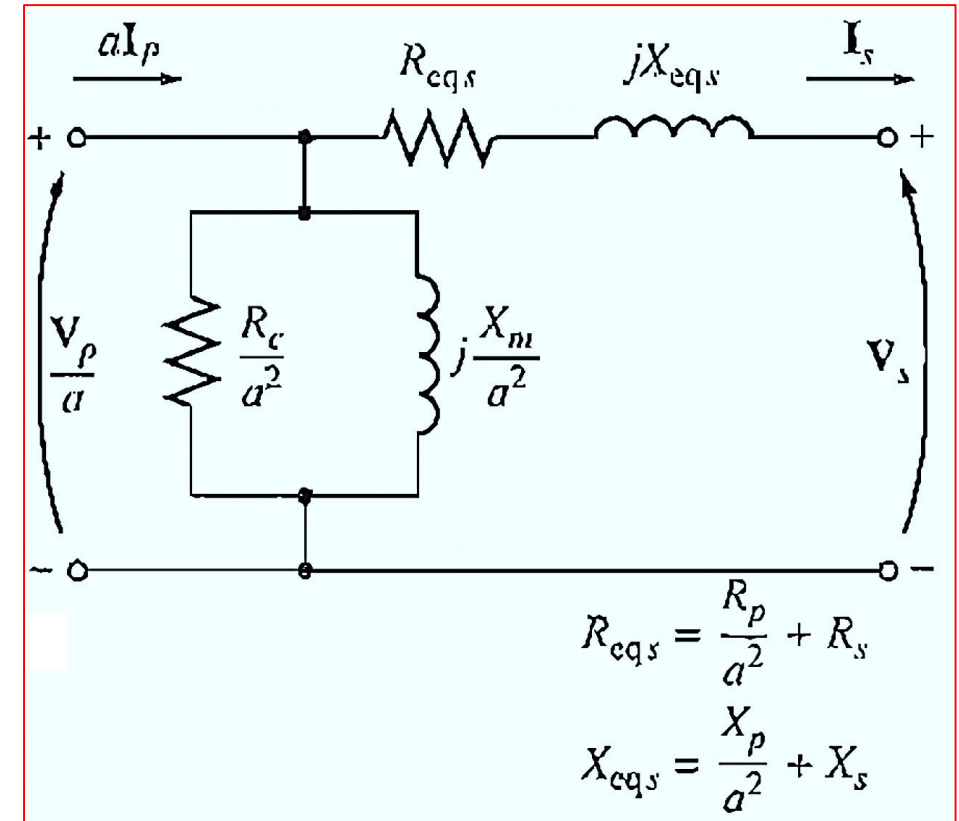


Transformer Phasor Diagram

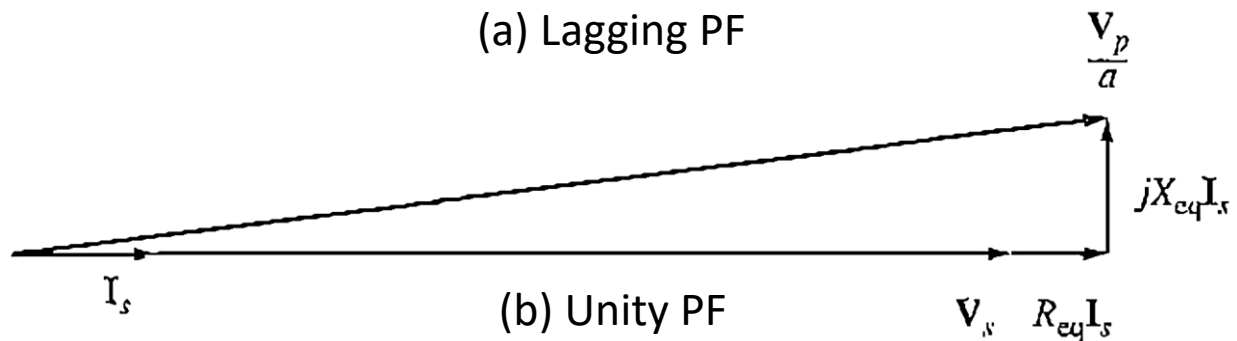
In all the following phasor diagrams, the phasor voltage V_s is assumed to be at an angle of 0° , and all other voltages and currents are compared to that reference. By applying Kirchhoff's voltage law to the equivalent circuit in the right figure, the primary voltage can be found as

$$\frac{V_p}{a} = V_s + R_{eq} I_s + jX_{eq} I_s$$

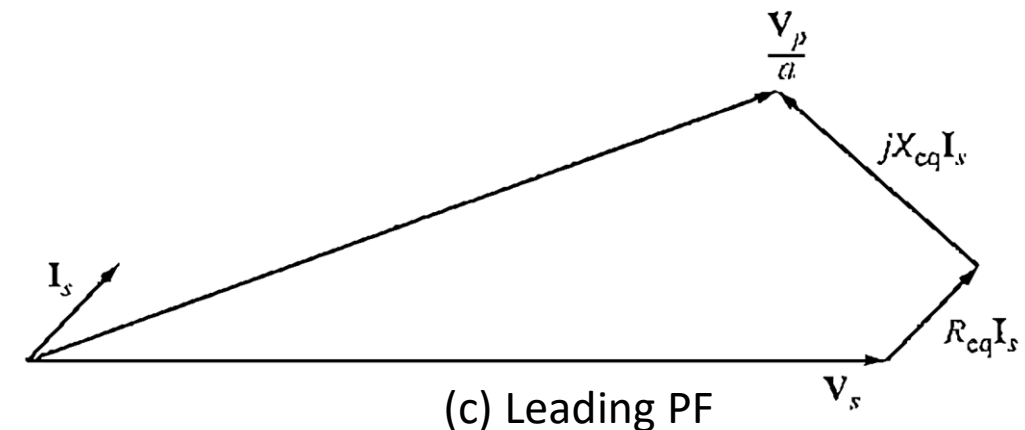
A transformer phasor diagram is just a visual representation of this equation.



(a) Lagging PF



(b) Unity PF



(c) Leading PF

Transformer Efficiency

The efficiency of a device is defined by the equation

$$\eta = \frac{P_{\text{out}}}{P_{\text{in}}} \times 100\%$$

$$\eta = \frac{P_{\text{out}}}{P_{\text{out}} + P_{\text{loss}}} \times 100\%$$

These equations apply to motors and generators as well as to transformers.

The transformer equivalent circuits make efficiency calculations easy. There are three types of losses present in transformers: (a) Copper (I^2R) losses, (b) Hysteresis loss, (c) Eddy current loss.

To calculate the efficiency of a transformer at a given load, just add the losses from each resistor and apply the equation below. Since the output power is given by

$$P_{\text{out}} = V_S I_S \cos \theta_S$$

the efficiency of the transformer can be expressed by

$$\eta = \frac{V_S I_S \cos \theta}{P_{\text{Cu}} + P_{\text{core}} + V_S I_S \cos \theta} \times 100\%$$

Example 2-5. A 15-kVA, 2300/230-V transformer is to be tested to determine its excitation branch components, its series impedances, and its voltage regulation. The following test data is given in the table.

- (a) Find the equivalent circuit of this transformer referred to the high-voltage side.
- (b) Find the equivalent circuit of this transformer referred to the low-voltage side.
- (c) Calculate the full-load voltage regulation at 0.8 lagging power factor, 1.0 power factor, and at 0.8 leading power factor using the exact equation for V_p .
- (d) Plot the voltage regulation as load is increased from no load to full load at power factors of 0.8 lagging, 1.0, and 0.8 leading.
- (e) What is the efficiency of the transformer at full load with a power factor of 0.8 lagging?

Open-circuit test (low voltage side)	Short-circuit test (high voltage side)
$V_{OC} = 230 \text{ V}$	$V_{SC} = 47 \text{ V}$
$I_{OC} = 2.1 \text{ A}$	$I_{SC} = 6.0 \text{ A}$
$P_{OC} = 50 \text{ W}$	$P_{SC} = 160 \text{ W}$

Solution

- (a) The turns ratio of this transformer is $a = 2300/230 = 10$. The excitation branch values of the transformer equivalent circuit referred to the secondary (low voltage) side can be calculated from the *open-circuit test* data, and the series elements referred to the primary (high voltage) side can be calculated from the *short-circuit test* data. From the open-circuit test data, the open-circuit impedance angle is

$$\theta_{OC} = \cos^{-1} \frac{P_{OC}}{V_{OC} I_{OC}}$$

$$\theta_{OC} = \cos^{-1} \frac{50 \text{ W}}{(230 \text{ V})(2.1 \text{ A})} = 84^\circ$$

The excitation admittance is thus

$$Y_E = \frac{I_{OC}}{V_{OC}} \angle -84^\circ$$

$$Y_E = \frac{2.1 \text{ A}}{230 \text{ V}} \angle -84^\circ \text{ S}$$

$$Y_E = 0.00913 \angle -84^\circ \text{ S} = 0.000954 - j0.00908 \text{ S}$$

The elements of the excitation branch referred to the secondary are

$$R_{C,S} = \frac{1}{0.000954} = 1050 \Omega$$

$$X_{M,S} = \frac{1}{0.00908} = 110 \Omega$$

From the short-circuit test data, the short-circuit impedance angle is

$$\theta_{SC} = \cos^{-1} \frac{P_{SC}}{V_{SC} I_{SC}}$$

$$\theta_{SC} = \cos^{-1} \frac{160 \text{ W}}{(47 \text{ V})(6 \text{ A})} = 55.4^\circ$$

The equivalent series impedance is thus

$$Z_{SE} = \frac{V_{SC}}{I_{SC}} \angle \theta_{SC}$$

$$Z_{SE} = \frac{47 \text{ V}}{6 \text{ A}} \angle 55.4^\circ \Omega$$

$$Z_{SE} = 7.833 \angle 55.4^\circ = 4.45 + j6.45 \Omega$$

The series elements referred to the primary side are

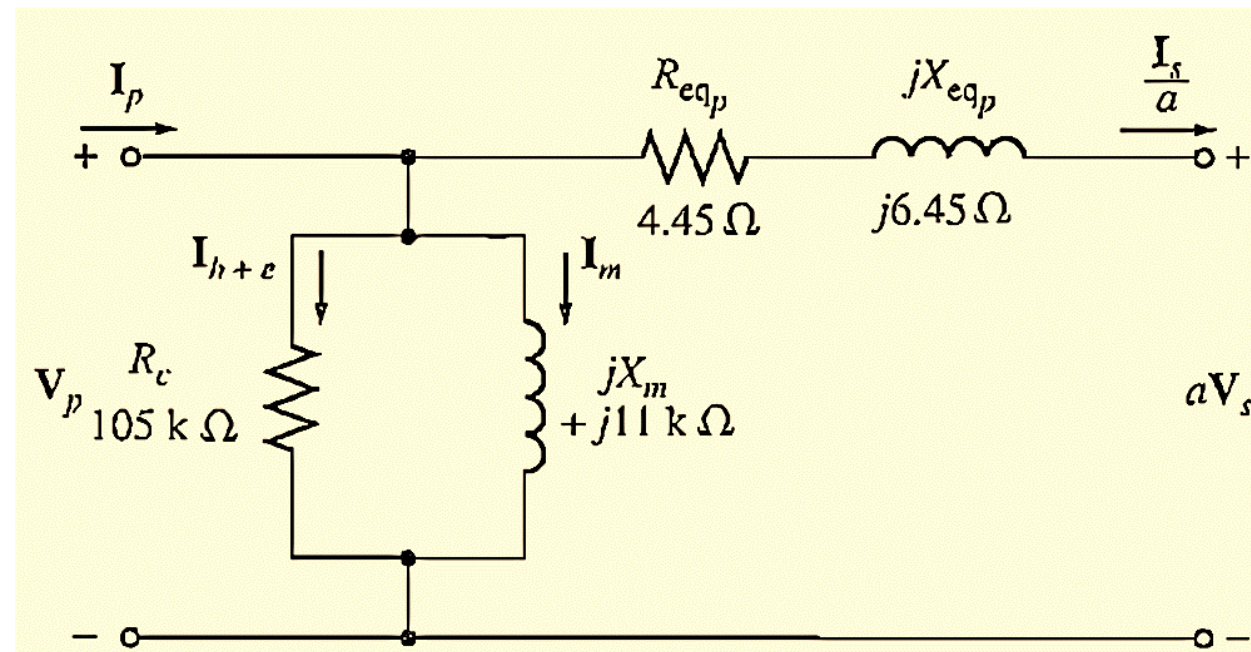
$$R_{eq,P} = 4.45 \Omega \quad X_{eq,P} = 6.45 \Omega$$

The resulting simplified equivalent circuit referred to the primary side can be found by converting the excitation branch values to the primary side.

$$R_{C,P} = a^2 R_{C,S} = (10)^2 (1050 \Omega) = 105 \text{ k}\Omega$$

$$X_{M,P} = a^2 X_{M,S} = (10)^2 (110 \Omega) = 11 \text{ k}\Omega$$

The resulting equivalent circuit is shown below:

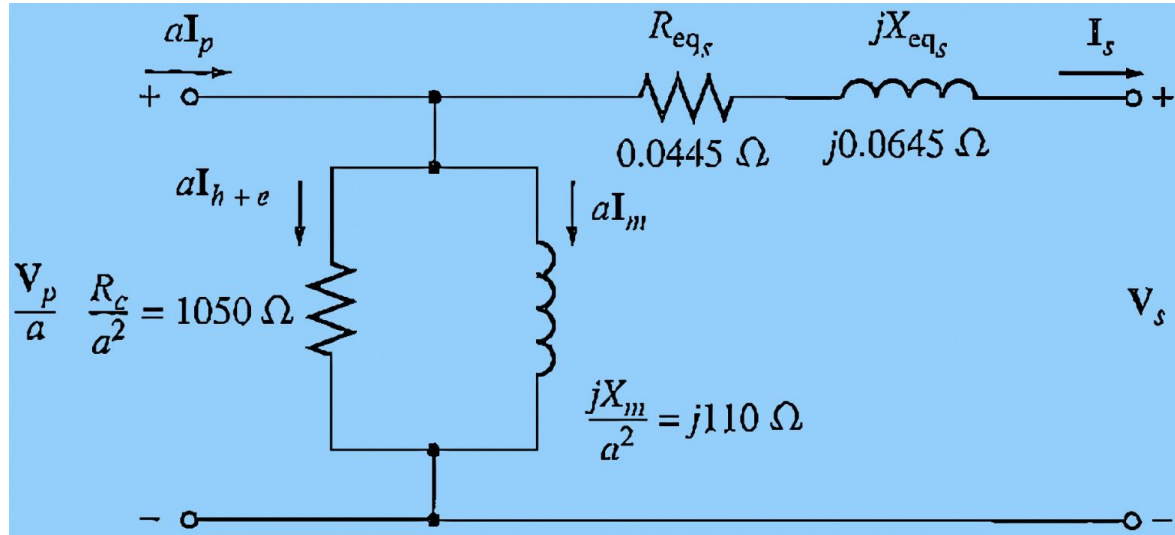


(b) To find the equivalent circuit referred to the low-voltage side, it is simply necessary to divide the impedance by a^2 . Since $a = N_p/N_s = 10$, the resulting values are

$$R_C = 1050 \, \Omega \quad R_{eq} = 0.0445 \, \Omega$$

$$X_M = 110 \, \Omega \quad X_{eq} = 0.0645 \, \Omega$$

The resulting equivalent circuit is shown below:



(c) The full-load current on the secondary side of this transformer is

$$I_{S, \text{rated}} = \frac{S_{\text{rated}}}{V_{S, \text{rated}}} = \frac{15,000 \text{ VA}}{230 \text{ V}} = 65.2 \text{ A}$$

To calculate V_p/a , use Equation (2-64):

$$\frac{V_p}{a} = V_s + R_{eq} I_s + jX_{eq} I_s$$

At PF = 0.8 lagging, current $I_s = 65.2 \angle -36.9^\circ \text{ A}$. Therefore,

$$\begin{aligned} \frac{V_p}{a} &= 230 \angle 0^\circ \text{ V} + (0.0445 \, \Omega)(65.2 \angle -36.9^\circ \text{ A}) + j(0.0645 \, \Omega)(65.2 \angle -36.9^\circ \text{ A}) \\ &= 230 \angle 0^\circ \text{ V} + 2.90 \angle -36.9^\circ \text{ V} + 4.21 \angle 53.1^\circ \text{ V} \\ &= 230 + 2.32 - j1.74 + 2.52 + j3.36 \\ &= 234.84 + j1.62 = 234.85 \angle 0.40^\circ \text{ V} \end{aligned}$$

The resulting voltage regulation is

$$\begin{aligned} \text{VR} &= \frac{V_p/a - V_{s, \text{fl}}}{V_{s, \text{fl}}} \times 100\% \\ &= \frac{234.85 \text{ V} - 230 \text{ V}}{230 \text{ V}} \times 100\% = 2.1\% \end{aligned}$$

At PF = 1.0, current $I_s = 65.2 \angle 0^\circ \text{ A}$. Therefore,

$$\begin{aligned} \frac{V_p}{a} &= 230 \angle 0^\circ \text{ V} + (0.0445 \, \Omega)(65.2 \angle 0^\circ \text{ A}) + j(0.0645 \, \Omega)(65.2 \angle 0^\circ \text{ A}) \\ &= 230 \angle 0^\circ \text{ V} + 2.90 \angle 0^\circ \text{ V} + 4.21 \angle 90^\circ \text{ V} \\ &= 230 + 2.90 + j4.21 \\ &= 232.9 + j4.21 = 232.94 \angle 1.04^\circ \text{ V} \end{aligned}$$

The resulting voltage regulation is

$$\text{VR} = \frac{232.94 \text{ V} - 230 \text{ V}}{230 \text{ V}} \times 100\% = 1.28\%$$

At PF = 0.8 leading, current $I_s = 65.2 \angle 36.9^\circ \text{ A}$. Therefore,

$$\begin{aligned} \frac{V_p}{a} &= 230 \angle 0^\circ \text{ V} + (0.0445 \, \Omega)(65.2 \angle 36.9^\circ \text{ A}) + j(0.0645 \, \Omega)(65.2 \angle 36.9^\circ \text{ A}) \\ &= 230 \angle 0^\circ \text{ V} + 2.90 \angle 36.9^\circ \text{ V} + 4.21 \angle 126.9^\circ \text{ V} \\ &= 230 + 2.32 + j1.74 - 2.52 + j3.36 \\ &= 229.80 + j5.10 = 229.85 \angle 1.27^\circ \text{ V} \end{aligned}$$

The resulting voltage regulation is

$$\text{VR} = \frac{229.85 \text{ V} - 230 \text{ V}}{230 \text{ V}} \times 100\% = -0.062\%$$

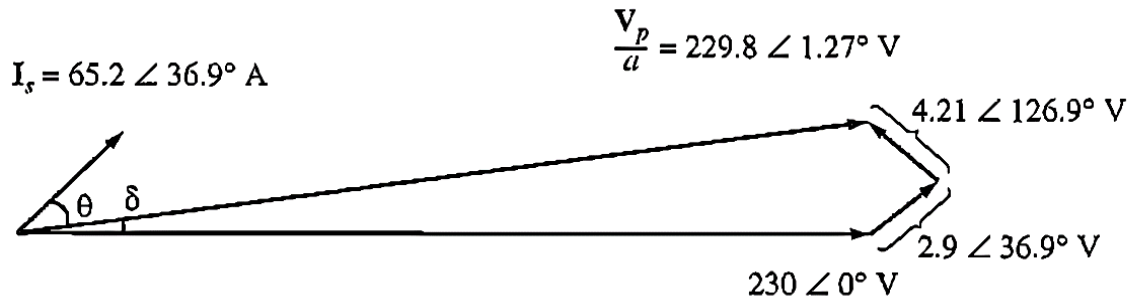
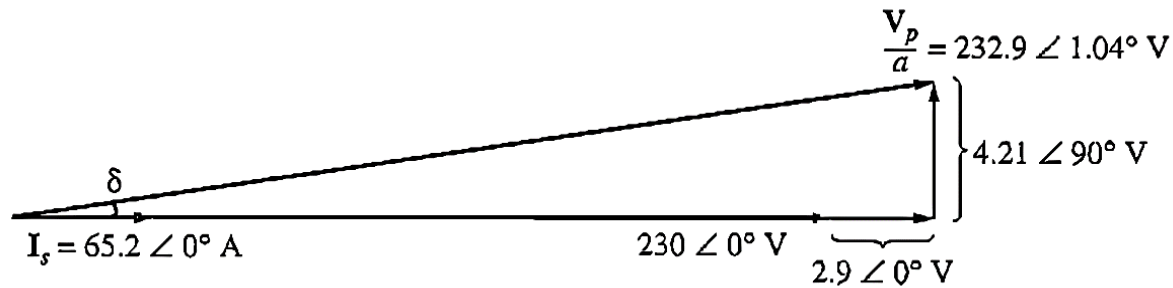
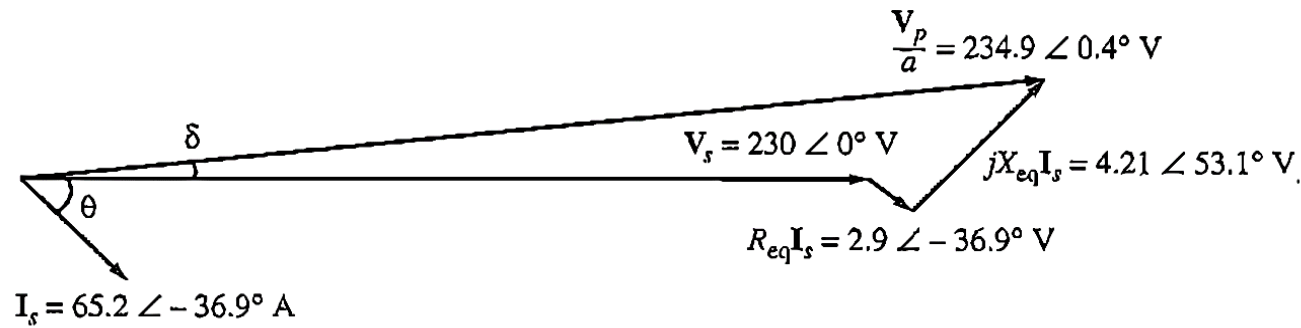
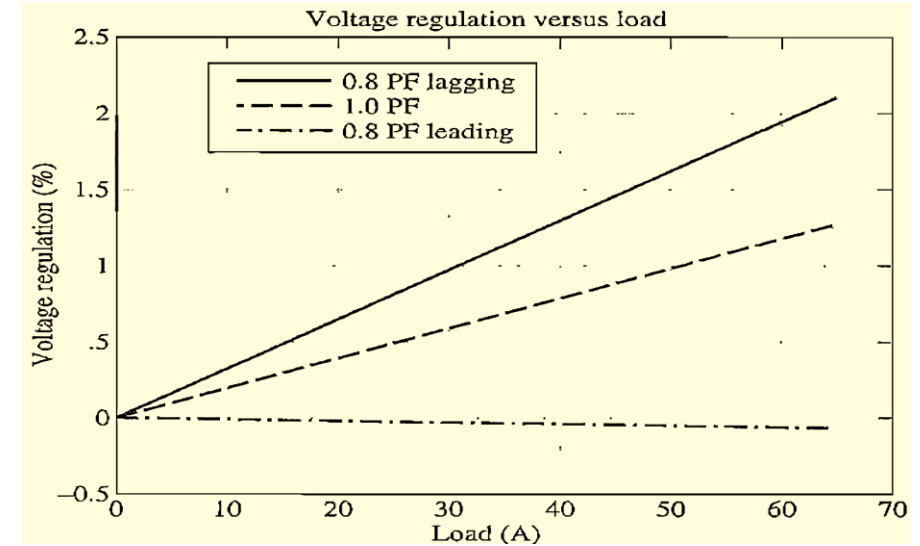


Fig. Phasor diagrams for Example 2-5 (b)

(d)



(e) To find the efficiency of the transformer, first calculate its losses. The copper losses are

$$P_{\text{Clt}} = (I_s)^2 R_{\text{eq}} = (65.2 \text{ A})^2 (0.0445 \Omega) = 189 \text{ W}$$

The core losses are given by

$$P_{\text{core}} = \frac{(V_p/a)^2}{R_C} = \frac{(234.85 \text{ V})^2}{1050 \Omega} = 52.5 \text{ W}$$

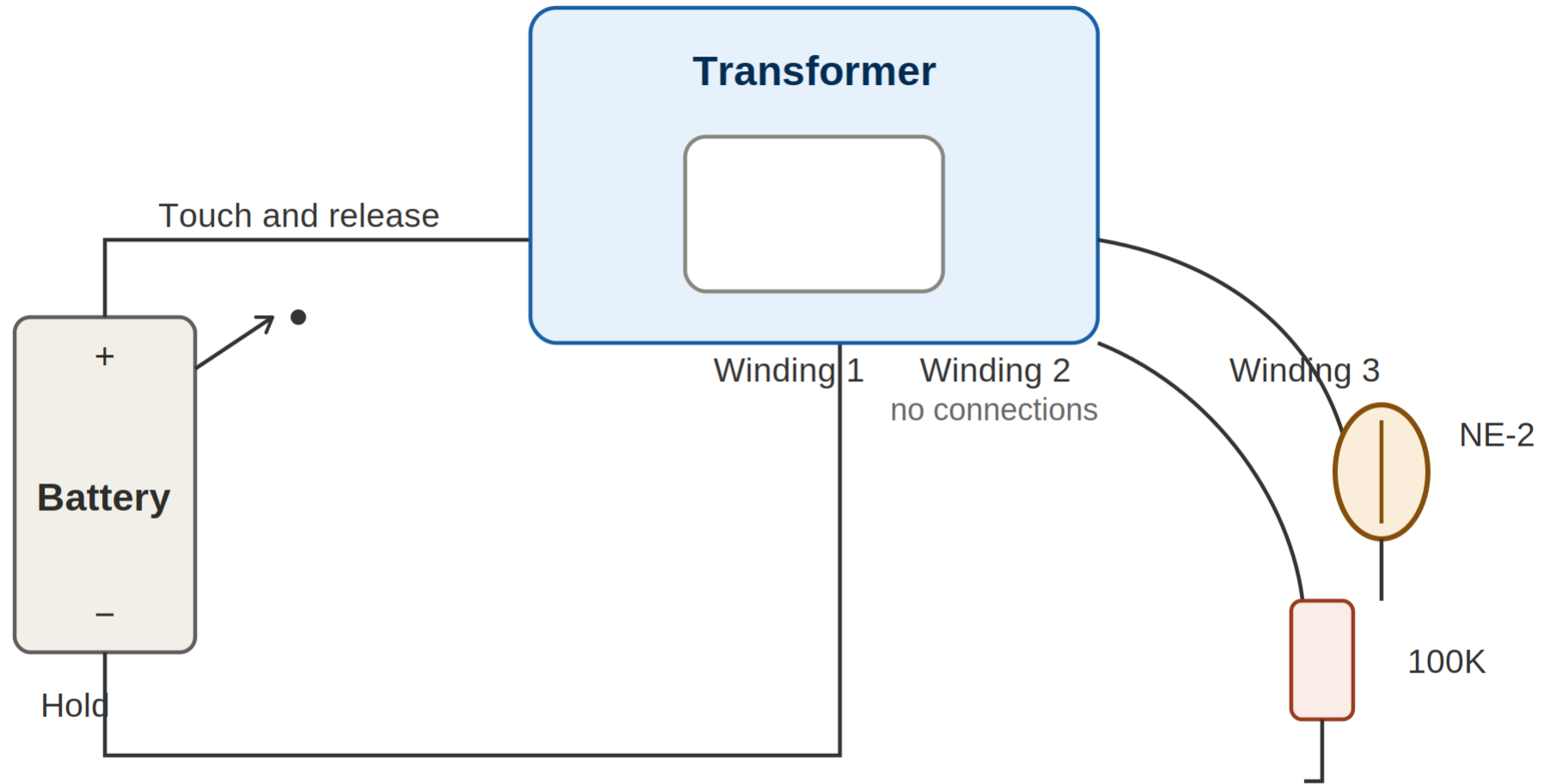
The output power of the transformer at this power factor is

$$\begin{aligned} P_{\text{out}} &= V_s I_s \cos \theta \\ &= (230 \text{ V})(65.2 \text{ A}) \cos 36.9^\circ = 12,000 \text{ W} \end{aligned}$$

Therefore, the efficiency of the transformer at this condition is

$$\begin{aligned} \eta &= \frac{V_s I_s \cos \theta}{P_{\text{Cu}} + P_{\text{core}} + V_s I_s \cos \theta} \times 100\% \\ &= \frac{12,000 \text{ W}}{189 \text{ W} + 52.5 \text{ W} + 12,000 \text{ W}} \times 100\% \\ &= 98.03\% \end{aligned}$$

TRANSFORMER ON DC SUPPLY



Lamination???

What will happen if the Primary of a Transformer
is Connected to D.C. Supply?

Transformer Taps & Voltage Regulation

- ♣ Distribution transformers have a series of taps in the windings to permit small changes in the turns ratio of the transformer.
- ♣ A typical installation provides for adjustments up to 5 percent above or below the nominal voltage rating of the transformer.
- ♣ The taps on a transformer permit the transformer to be adjusted in the field to accommodate variations in local voltages.
- ♣ The taps on a transformer permit the transformer to be adjusted in the field to accommodate variations in local voltages . However, these taps normally cannot be changed while power is being applied to the transformer. They must be set once and left alone.

❖ *How can a power company supply a controlled voltage through high-impedance lines to loads which are constantly changing?*

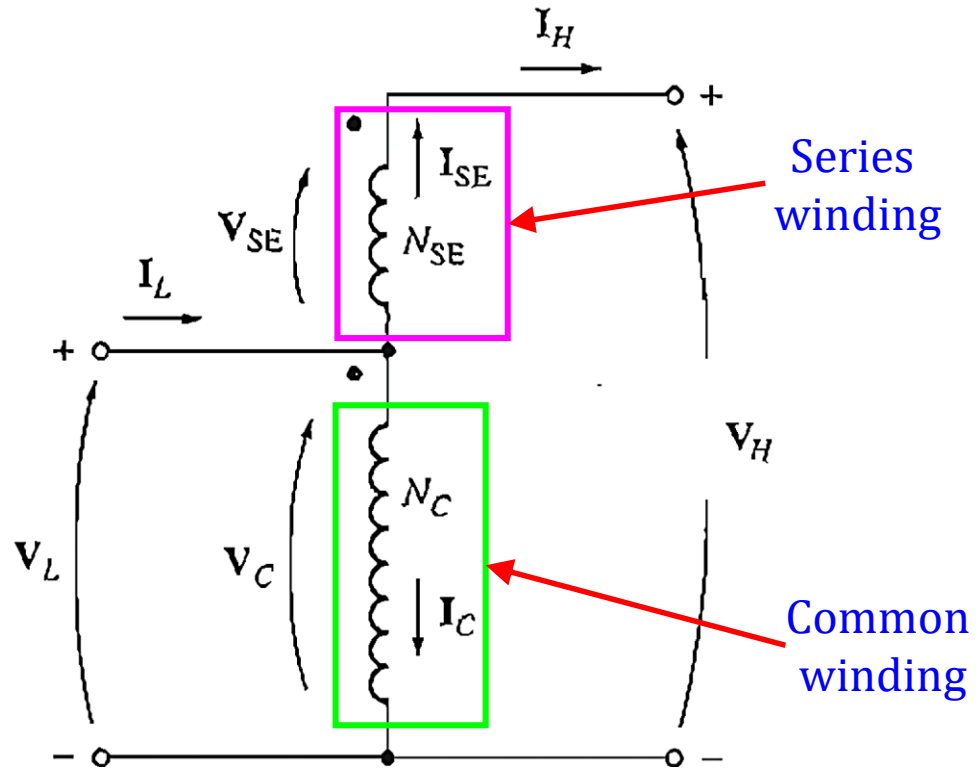
One solution to this problem is to use a special transformer called a tap changing under load (TCUL) transformer or voltage regulator. Basically, a TCUL transformer is a transformer with the ability to change taps while power is connected to it. A voltage regulator is a TCUL transformer with built-in voltage sensing circuitry that automatically changes taps to keep the system voltage constant.

❖ *Example 2-6. A 500-kVA, 13,200/480-V distribution transformer has four 2.5 percent taps on its primary winding. What are the voltage ratios of this transformer at each tap setting?*

+5.0% tap	13,860/480 V
+2.5% tap	13,530/480 V
Nominal rating	13,200/480 V
- 2.5% tap	12,870/480 V
-5.0% tap	12,540/480 V

The Autotransformer

Autotransformer is a special purpose transformer used for small range voltage conversion where a single coil is used as both primary and secondary winding.



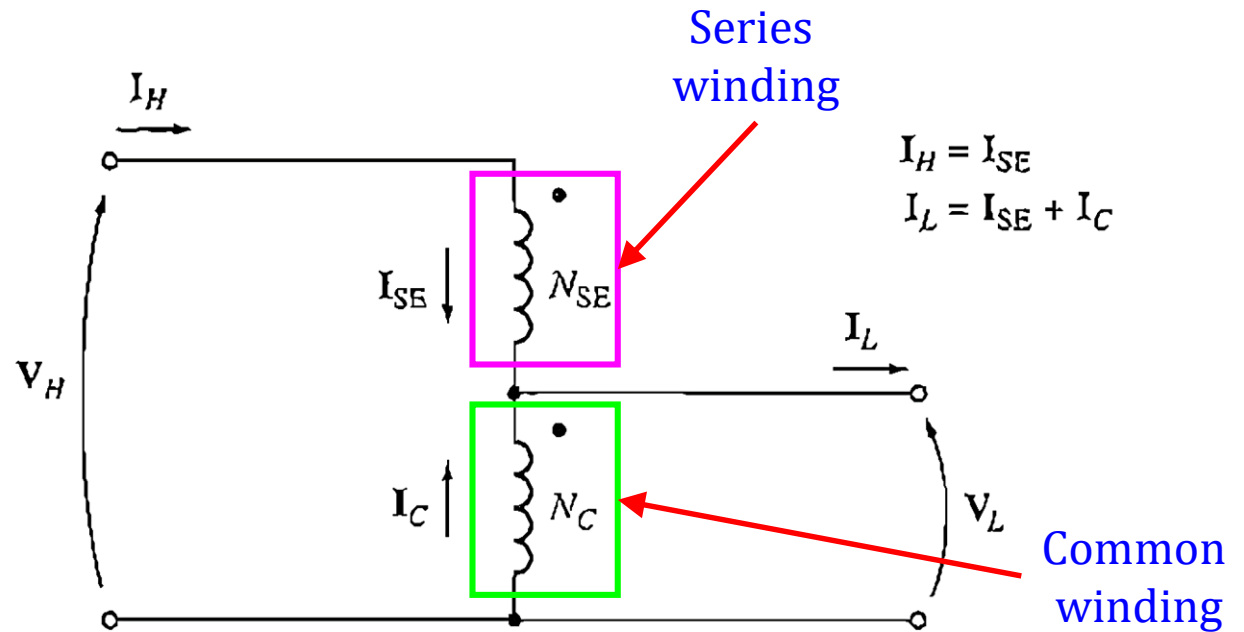
Step up autotransformer

$$\frac{V_C}{V_{SE}} = \frac{N_C}{N_{SE}}$$

$$N_C I_C = N_{SE} I_{SE}$$

$$V_L = V_C$$

$$V_H = V_C + V_{SE}$$



Step down autotransformer

$$I_L = I_C + I_{SE}$$

$$I_H = I_{SE}$$